

FOREST ENTOMOLOGY LAB: BRANCH-TO-BENCH USER MANUAL

VERSION 3.0

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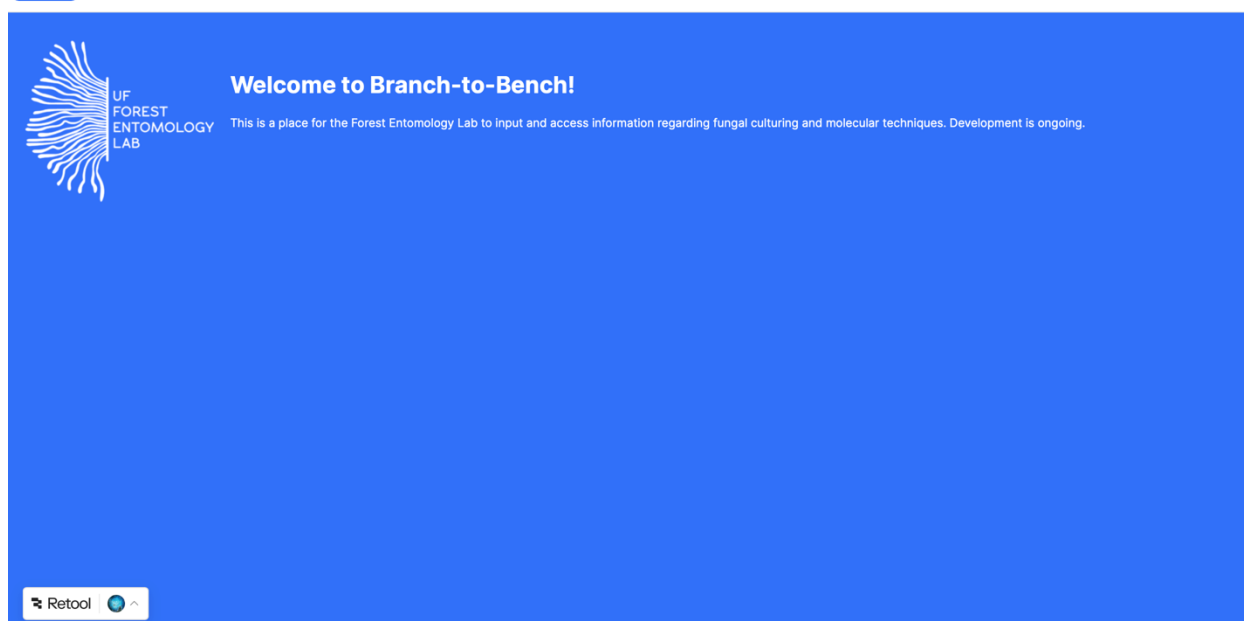
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ACCESS THE DATABASE

To access the molecular database application, go to this link (<https://forestentomology.retool.com/apps/a7db05a4-5a73-11ef-8429-cfa94442fbd9/Version%203.0/Home%20Page%203.0>), and log in using the username **forestentomology@ifas.ufl.edu** and the password **Ambrosia1!**.

HOME PAGE

This acts as a landing page for the database. There are no options for data visualization or input here, but there is a navigation bar at the top of the page that permits users to move to the Projects, People, Beetles, Media Prep, Cultures, Slants, Extractions, PCR, Sequencing, Processes, and Sample Tracking applications. This navigation bar appears at the top of all pages.



PROJECTS PAGE

This page contains information on projects. Almost all objects in the database are linked to a project. This helps govern information to be filtered and sorts samples. Additionally, it represents an additional layer of reinforcement of meaning. Ideally, terms should be used in similar ways across projects, but at the very least, terms referring to samples *within* a project should be consistent. There are three characteristics associated with a project. Firstly, there is a *title* applied to the project, which is how it is referenced elsewhere in the database. Secondly, there is a short *description*, which gives additional detail for future users as to its purpose and parameters. Lastly, there is a *status* that indicates whether the project is active (people are currently working on it and generating samples), awaiting publication (people are no longer generating samples but the paper has yet to be published), published (for old projects where people are no longer generating samples but there exists a corresponding paper or publication), suspended (for projects where the intent is to resume them at some later time but samples are not currently being generated), and abandoned (for projects that do not have a publication associated with them, are not currently generating data, and for which residual samples may no longer be needed). There are two purposes for these categories. The first is related to the user interfaces on other pages: projects not marked as “active” will not show up as options elsewhere. Secondly, when cleaning out the lab, being able to determine which projects are in various stages of completion will assist with determining whether certain samples can be safely removed.

In many cases, laboratories will start new experiments with samples derived from older projects (ex: starting a culture from a frozen slant) rather than generating a fresh set of material. You do not need to reassign these objects’ project identities to retain full database functionality. Although the single-entry forms on each page often initially suggest source samples restricted to a particular project to make data input easier, it is possible to designate any eligible object from any project as a source in the review tables associated with bulk entry on each page. Similarly, the sample tracking application will automatically “grab” any objects from which samples in the selected project were derived, regardless of whether they also belong to that project.

Thus, users have flexibility in terms of how broadly or narrowly they sort samples into “projects” without needing to worry about losing information at the final stage of data collation.

“LOGGING IN”

At the bottom of every application page with the exception of “Home” and “Sample Tracking”, there is a “Log In to Edit” box inviting the user to provide a username and password. Because on the free Retool plan, there is no way to restrict user’s access to the pages or to the Retool Resources where our data is stored, this login is less for security purposes and more to prevent users from editing or deleting information that other people have input into the database. (If you want a more secure way to do this or implement multiple users, we suggest subscribing to a paid account through their platformed.) Your ability to add additional information to the database is also restricted when you are not logged in. However, you can still examine the tables and use the filtering options to adjust your views.

Log In To Edit

Username

Password



Submit

To log in, input your username and password into the form provided and click submit. Once you click submit, the form for inputting a new project (or on other pages, inputting other types of new information into the database) will appear automatically. You will also be greeted with a small band of text at the bottom of the page advising you as to who you are logged in as and to refresh if you want to log in and edit as someone else.

You're editing entries as *Miranda Barnes*. If you want to switch users, please refresh the page.

USING THE TABLE

The dimensions of the columns in the Projects table can be altered by clicking and dragging the column edges at the top of the table where the headings sit. Click a second time to release the column edge into its

new position.

Projects Dashboard	
Title	Description
Euplatypus Project (Dummy Data)	Old data associated with this paper (https://link.springer.com/article/10.1007/s13199-023-00938-4) which M
General Lab Use	Not associated with a specific project, for general use
Platypodine Diversity	Project seeking to clarify features of Platypodinae and their fungi
SPB Fungal Symbiont Genome	Sequencing Entomocorticium and Ceratocystiopsis from the Southern pine beetle (<i>Dendroctonus frontalis</i>)
SPB Genome	Sequencing the genome of the southern pine beetle (<i>Dendroctonus frontalis</i>)
SPB Mycangial Transcriptome	Samples associated with generating transcriptome data and differential gene expression analysis for the mycan
Wingfield Sabbatical	Samples associated with Mike and Brenda Wingfield coming on sabbatical to the lab
Myrmecine Genome	Sequencing the genome of the myrmecine beetle (<i>Myrmecodius</i>)
9 results	
Filtering Options	

Items in the Projects Dashboard can be filtered by toggling the “Filtering Options” container open. You have three options for fields to filter by: “Title”, “Description”, and “Status”. “Title” and “Description” are both filtered by entering text into their respective boxes. If you want to filter for multiple terms at once (i.e., to select projects that include EITHER the terms “SPB” and “Sabbatical”), separate each search term with a semicolon followed by a space (i.e., “SPB; Sabbatical”). For “Status”, you may check or uncheck as many boxes of the multiselect menu as you would like to influence what is displayed in the table above.

Filtering Options	
Title	Status
Separate different search terms with semicolon-space	Abandoned ✓
Description	Active ✓
Separate different search terms with semicolon-space	Awaiting Publication ✓

All fields in the Projects table can be edited. To do this, log in and click on the desired field. A text cursor and box highlight will appear in Title and Description fields, and a dropdown menu will appear beneath the selected box under “Status”. For the text entries, the user must click out of the box for the table to register that an edit has been made.

Title	Description
Euplatypus Project (Dummy Data)	Old data associated with this paper (https://link.springer.com/article/10.1007/s13199-023-00938-4) which Miranda
General Lab Use	Not associated with a specific project, for general use
Platypodine Diversity	Project seeking to clarify features of Platypodinae and their fungi
SPB Fungal Symbiont Genome	Sequencing Entomocorticium and Ceratocystiopsis from the Southern pine beetle (Dendroctonus frontalis)
SPB Genome	Sequencing the genome of the southern pine beetle (Dendroctonus frontalis)
SPB Mycangial Transcriptome	Samples associated with generating transcriptome data and differential gene expression analysis for the mycangium
Wingfield Sabbatical	Samples associated with Mike and Brenda Wingfield coming on sabbatical to the lab
Yucca Genome	Sequencing the genome of the granulate ambrosia beetle (Y. crassiusculus)

9 results

	Status
ciated with this paper (https://link.springer.com/article/10.1007/s13199-023-00938-4) which Miranda is using as dummy data to test database fun...	Published
d with a specific project, for general use	
g to clarify features of Platypodinae and their fungi	
ntomocorticium and Ceratocystiopsis from the Southern pine beetle (Dendroctonus frontalis)	
ie genome of the southern pine beetle (Dendroctonus frontalis)	
ciated with generating transcriptome data and differential gene expression analysis for the mycangium of the southern pine beetle	
ciated with Mike and Brenda Wingfield coming on sabbatical to the lab	
ie genome of the granulate ambrosia beetle (Y. crassiusculus)	

9 results

Search...

- Abandoned
- Active
- Awaiting Publication
- Published
- Suspended

Once a selection or text edit is complete, the edited area will be flagged with a blue triangle, and buttons to cancel or save the edits will appear at the bottom of the table. Clicking “Cancel” will undo the edits in the table, and clicking “Save” will commit the edits to the database. **If the user does not click “Save” before leaving the page, any changes made to the table will not be stored in the database, and the edits will not appear when the user returns to “Projects” or other pages in the application.**

	Status
ciated with this paper (https://link.springer.com/article/10.1007/s13199-023-00938-4) which Miranda is using as dummy data to test database fun...	Awa...
d with a specific project, for general use	Acti...
g to clarify features of Platypodinae and their fungi	Acti...
ntomocorticium and Ceratocystiopsis from the Southern pine beetle (Dendroctonus frontalis)	Acti...
ie genome of the southern pine beetle (Dendroctonus frontalis)	Acti...
ciated with generating transcriptome data and differential gene expression analysis for the mycangium of the southern pine beetle (Dendroctonus fr...	Acti...
ciated with Mike and Brenda Wingfield coming on sabbatical to the	Acti...
ie genome of the granulate ambrosia beetle (Y. crassiusculus)	Acti...

1 change Cancel Save

9 results

Projects can also be deleted. To delete a project, click the box to the left of the Title field. This will cause a ‘Delete Selected Projects’ button to appear at the bottom of the page. The user can select as many projects

as they wish to delete before clicking “Delete Selected Projects”.

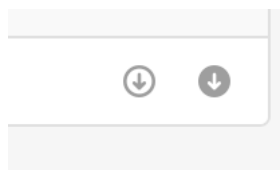
 Title	Description
☒ Euplatypus Project (Dummy Data)	Old data associated with this paper (https://link.springer.com/article/10.1007/s13199-023-00938-4) which M
General Lab Use	Not associated with a specific project, for general use
Platypodine Diversity	Project seeking to clarify features of Platypodinae and their fungi
SPB Fungal Symbiont Genome	Sequencing Entomocorticium and Ceratocystiopsis from the Southern pine beetle (Dendroctonus frontalis)
SPB Genome	Sequencing the genome of the southern pine beetle (Dendroctonus frontalis)
SPB Mycangial Transcriptome	Samples associated with generating transcriptome data and differential gene expression analysis for the mycan
Wingfield Sabbatical	Samples associated with Mike and Brenda Wingfield coming on sabbatical to the lab
X. crass Genome	Sequencing the genome of the granulate ambrosia beetle (X. crassiusculus)
1 of 9 selected  	

Delete Selected Project(s)

When the user selects “Delete Selected Project(s)”, they will be prompted a second time to confirm whether or not they wish to delete that entry. If the user selects “No, do not delete.”, the prompt will disappear and the item will not be deleted. If the user selects “Yes, I’m sure.” the query to delete the project will be submitted and it will be removed from the database.

Projects can only be deleted if there are no samples associated with them. If a user accidentally selects a project that falls into this category, pressing the delete button will not remove it – it will only remove the projects without samples in the user selection. Later in the manual, you will see this will also applies to samples that are not generated by the person who is currently logged in. However, because there is no member attribution in the projects, anyone can add and delete any isolated project.

There are two options for downloading information in the “Projects” table. Clicking the download button (circle with an arrow pointing downward) with the dark-colored arrow at the bottom right-hand-corner of the display gives the user a raw representation of the information in the table – i.e., the “Status” column of the resulting .csv file is filled with numbers representing the corresponding entries for those terms in the database.



A1 Title		
Name Box		
	A	B
1	Title	Description
2	General Lab Use	Not associated with a specific project, for general u
3	Platypodine Diversity	Project seeking to clarify features of Platypodinae a
4	SPB Fungal Symbiont Genome	Sequencing Entomocorticiu and Ceratocystiopsis
5	SPB Genome	Sequencing the genome of the southern pine beetle
6	Wingfield Sabbatical	Samples associated with Mike and Brenda Wingfield
7	X. crassiusculus Fungal Symbio	Sequencing Ambrosiella roeperi from Xylosandrus c
8	Xcrass Genome	Sequencing the genome of the granulate ambrosia b

The download button at the bottom right-hand-corner with the white arrow will download a human-readable “mapped” .csv version of this file.

A1 Title		
Name Box		
	A	B
1	Title	Description
2	General Lab Use	Not associated with a specific project, for general
3	Platypodine Diversity	Project seeking to clarify features of Platypodinae
4	SPB Fungal Symbiont Genome	Sequencing Entomocorticiu and Ceratocystiops
5	SPB Genome	Sequencing the genome of the southern pine beetl
6	Wingfield Sabbatical	Samples associated with Mike and Brenda Wingfie
7	X. crass Genome	Sequencing the genome of the granulate ambrosia
8	X. crassiusculus Fungal Symbiont	Sequencing Ambrosiella roeperi from Xylosandrus

CREATING A NEW PROJECT

To create a new project, fill out the “New Project” form by typing in the desired title and description and then clicking Submit.

New Project

Title *

Short nickname for project

Description *

The tasks, goals, etc. associated with this project

Submit

If a user attempts to name a project a title that already exists, the form will not submit.

New Project

Title *

X. crass Genome

This project already exists.

Description *

The tasks, goals, etc. associated with this project

Submit

Both fields are mandatory. If they are not filled, the form will not submit.

All project submissions are automatically recorded as “Active”. If this is not the desired status for a given project, then the user can alter it in the table after it has been submitted.

PROJECT TABLE TERMS

The last item on this page is a table full of terms relating to “Projects” (i.e., the names of attributes and definitions for the statuses). This table can be filtered according to the same principles as the “Projects” table itself.

Table Terms	
Term	Definition
Descripti...	A more in-depth explanation of the project's objectives
Published	This project is complete and published
Status	How the project is currently classified according to whether it is actively being worked on, on hold, published, etc.
Suspended	This project is not complete or presently ongoing, but will be resumed at a future time
Title	A short phrase descriptive of the project that appears with associated objects in tables elsewhere in the user interface
8 results	
Filtering Options	
Term	Definition
Separate different search terms with semicolon-space	Separate different search terms with semicolon-space

PEOPLE PAGE

This page contains information on lab members associated with various samples. Before entering any new samples, users will need to input information associated with the person who created their sample into this table. Almost all objects in the database are linked to a creator. This helps when filtering information and sorting samples. There are four characteristics associated with a person/lab member. There is a *First Name*, *Last Name*, and *Role* describing the individual’s position in the laboratory. Lastly, there is a *Status* that indicates whether the person is “Active (Employed)”, “Active (Visiting)” or “Inactive” (i.e., no longer working in the lab, even if they may still have samples recorded in the database.) People not marked as “active” will not show up as options on the other pages.

Users must log in before they are able to add new members or change their passwords. The process of logging in is as described for “Projects”.

USING THE TABLE

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”.

People Dashboard			
Last Name	First Name	Role	Status
Barnes	Miranda	MS Student	Active (Employed)
Deitz	Katy	PhD Student	Active (Employed)
Dong	Yiyi	Postdoc	Active (Employed)
Hulcr	Jiri	Principal Investigator	Active (Employed)
Johnson	Andrew	Scientist	Active (Employed)
Kelley	Daniel	Lab Manager	Active (Employed)
Kulbeda	Averie	Undergraduate Student	Active (Employed)
Marais	Christopher	PhD Student	Active (Employed)
13 results			 

Different fields can be filtered using the dropdown “Filtering Options” menu, as described in Projects. Filtering options are shown below.

Filtering Options

First Name

Separate different search terms with semicolon-space

Last Name

Separate different search terms with semicol

Role

Separate different search terms with semicolon-space

Status

Active (Employed) ✓

Active (Visiting) ✓

Inactive ✓

All fields in this table can be edited in the same manner as in Projects, with the same interface. “First Name”, “Last Name”, and “Role” are edited as text, and “Status” is edited through a dropdown menu. Rows can also be deleted in the same manner as in Projects. Again, rows will only be removed if they aren’t directly referenced by other samples in the database.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “People” table.

CREATING A NEW LAB MEMBER

To create a new lab member, fill out the “Add New Lab Member” form by typing in the desired “First Name”, “Last Name”, and “Role”; selecting a Status from the drop-down menu; and then clicking submit. All of these fields are required. Users may also optionally provide a username and password for the name input. If a particular member is expected to add or edit data in the database at some point in time, we recommend filling these fields (neither are required unless the user inputs text into one or the other). If individuals are being added primarily for recordkeeping purposes, these can be left empty.

Add New Lab Member

First Name *

Enter first name of lab member

Last Name *

Enter last name of lab member

Role *

Master's student, lab manager, etc.

Status *

Select an option

Username

Enter value

Password

Enter value

Submit

PEOPLE TABLE TERMS

Definitions for different statuses can be found in the “Table Terms” display at the bottom of the page. This table can be filtered and downloaded as with the one on the “Projects” page.

Table Terms	
Term	Definition
Active (Employed)	This lab member is actively employed in the Forest Entomology laboratory
Active (Visiting)	This person is participating in lab activities on a short-term basis
Inactive	This person has been involved in laboratory activities in the past but is no longer present
3 results	
Filtering Options	
Term	Definition
Separate different search terms with semicolon-space	Separate different search terms with semicolon-space

CHANGE PASSWORD

If a user is logged in on the “People” page, they can change their password by filling in the “Change Password” form at the bottom of the page and then clicking submit.

Change Password

New Password

Submit

BEETLES PAGE

The “Beetles” page is one of the more heavily populated interfaces in the database application, managing tables for storing information on “Beetles”, “Identifications” of those beetles, and potential sources for those beetles (i.e., “Trees”, “Galleries”, and “Traps”). As with the preceding pages, input forms are hidden unless login information is provided.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE BEETLES TABLE

The “Beetles” table includes fields for “Beetle ID” (displaying the unique identifier assigned to a given beetle, in Scolytos or otherwise), the “Taxon” (the most recent species identification for the beetle), the “Life Stage” (i.e., larva, pupa, or adult), the “Sex”, the “Sample Size” (the number of beetles encompassed by an ID), the “Sampling Protocol”, “Derived From” indicating (if applicable) which tree, gallery, or trap a beetle came from, “Vial Number” (reflecting the number of the vial the beetle is contained in, if recorded in Scolytos), the “Date Collected”, the person the beetle was “Collected By”, and the “Project” the beetle is associated with, as well as information on the location where the beetle was recorded (“Country”, “State/Province”, “County”, “Locality”, “Latitude”, “Longitude”) and any “Notes” the user might have on the sample.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in Projects. Concealed columns can be revealed by scrolling to the side.

Beetles Dashboard							
Beetle ID	Taxon	Life Sta...	Sex	Sample SI...	Sampling Protocol	Derived From	Vial... Date
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Crossotarsus wallacei	Adult	Male	1	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Crossotarsus wallacei	Adult	Male	1	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Crossotarsus wallacei	Adult	Female	1	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Crossotarsus wallacei	Adult	Female	1	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Ecoptopterus spinosus	Adult	Female	1	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Ambrosiodmus lewisi	Adult	Female	1	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Anisandrus ursa	Adult	Female	3	Field-sampled from galleries		Apr 15, 2025
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	Crossotarsus wallacei	Adult	Female	1	Field-sampled from galleries		Apr 15, 2025

Filtering Options

Beetle ID

Separate different search terms with semicolon-space

Taxon

Separate different search terms with semicolon-space

Life Stage

Adult

Larva

Pupa

Sex

Female

Mixed Sex

Not recorded

Male

Show samples with missing sex/life stage information

Sample Size

1-60

Sampling Protocol

Field-sampled from galleries

Flying or trapped

Light-trapped live

Live mycangial dissection from reared beetle

Derived From

Separate different search terms with semicolon-space

Vial Number

Separate values with semicolon-space

Date Range

Mar 24, 2006 - Feb 17, 2026

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Jiri Hulcr

Country

Separate with semicolon-space

County

Separate with semicolon-space

Latitude

-90-90

Longitude

-180-180

Notes

Separate different search terms with semicolon-space

Projects

Ambrosiella Phase Transition

Cycloheximide Concentration

Dinoplatypus Project

Euplatypus Project (Dummy Data)

State or Province

Separate with semicolon-space

Locality

Separate with semicolon-space

You can only edit beetles in the Beetles table associated with your lab member identity (i.e., the person listed in the “Collected By” column). The page determines whether your ID matches the ones in the table based on the login information provided at the bottom of the page. “Beetle ID”, “Country”, “State/Province”, “County”, and “Locality” may be edited as text. “Sample Size”, “Vial Number”, are similarly edited as text but only accept numerical input. “Life Stage”, “Sex”, “Sampling Protocol”, “Derived From”, and “Project” are edited through dropdown menus. “Date Collected” can be edited through a calendar menu. Rows can also be deleted in the same manner as in “Projects”. Note that you cannot edit “Taxon” directly in the beetle table. This is because these are all fields associated with a beetle identification that can only be modified via the “Make New Identifications” form.

Starting with the Beetles table and continuing through through most other tables discussed in this manual, if a table contains entries where the person who created an item is stored, individuals cannot delete entries unless they were the ones marked in the database as having created them. Ideally, the same person who collected the information should be the one managing it in the database. However, if these are not the same, put the data manager's name in the appropriate column, then indicate in the notes field that it was produced by someone else. Similarly to projects and people, rows will only be removed if there are no downstream samples that reference that beetle.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Beetles” table.

USING THE IDENTIFICATIONS TABLE

The “Identifications” table includes fields for the unique “Beetle ID” of the beetle displayed, the “Taxon” to which the beetle has been identified as belonging to, the “Date Identified”, the person it was “Identified By”, the “Identification Reference” they used (if applicable), and any “Notes” on the identification. This table exists separately from the “Beetles” table for a few reasons. Firstly, this allows multiple identifications to be

applied to a given beetle by different people at different times. This way, if there is an identification disagreement between two parties or one method of identification disagrees with another, conflicting information can be retained in the database. Secondly, identification does not always necessarily happen concurrently with sampling. Treating it as a separate “event” formalizes the act of identification, particularly when it occurs after the sample was generated.

To view the identifications assigned to different beetles, users click on the checkboxes beside the beetles they want to examine. The “Make New Identification” form will also automatically populate with the ID for the most recently selected beetle. Identifications can be deleted from the “Identifications” table in the same way that they can from the “Beetles” table (also restricted by user). However, it is recommended that users retain even erroneous identifications to maintain an accurate history of how a sample was treated over the course of its lifespan. You cannot edit any identifications entries for this reason – if you want to add a new taxon to the table for a sample, you must create a new entry.

Identifications

Taxon	Date Identified	Identified ...	Identification
Hulcr & Narakusumo Crossotarsus wallacei	Apr 17, 2025 2:29 PM	Jiri Hulcr	

1 result

Make New Identification

Beetle ID *
INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo 002

Taxon *
Species name

Life Stage *
Adult

Sex *
Male

Identification Reference
Source used to ID (optional)

Date Identified *
May 05, 2025 11:17...

Identified By *
Miranda Barnes

Notes
Any other relevant information

Submit

Identifications can be downloaded and filtered as with the “Projects” table using the buttons at the base of the display. Filtering options are shown below.

Filtering Options

Beetle ID
Separate different search terms with semicolon-space

Taxon
Separate different search terms with semicolon-space

Date Range
Jan 1, 2019 - May 6, 2025

Lab Member

- Miranda Barnes ✓
- Katy Deltz ✓
- Yiyi Dong ✓
- Jiri Hulcr ✓
- Andrew Johnson ✓

Identification Reference
Separate terms with semicolon-space

Notes
Separate terms with semicolon-space

CREATING A NEW IDENTIFICATION

To make a new identification, users must select a “Beetle ID” from the list of valid beetles, add a taxon, indicate the date identified, and record who identified the beetle. Users provide text input for “Taxon”, “Identification Reference”, and “Notes”. They may select from a dropdown menu for “Beetle ID”, “Life Stage”, “Sex”, and “Identified By”. To create a new identification, users must click submit. If you are modifying life stage or sex, you must fill both fields (even if you’re marking that the sex was not recorded). However, this shouldn’t come up very frequently since the database will generally autofill information from the selected row based on existing life stage/sex information and not leave either field empty.

CREATING A NEW BEETLE

There are two ways to add beetles to the database – singly or in bulk. To draft single entries, there is a “New Beetle Record Form.” Users provide text input for “Beetle ID”, “Country”, “State/Province”, “County”, “Locality”, “Notes”, “Taxon”, “Identification Reference”, and “[Identification] Notes” and select from a dropdown menu for “Associated Project”, “Source ID”, “Sampling Protocol”, “Date Collected”, “Collected By”, “Life Stage”, “Sex”, and “Identified By”. “Date Collected” and “Date Identified” ask for users to specify a date from a calendar selection. “Sample Size”, “Vial Number”, “Latitude”, and “Longitude” all accept text input, but only in numerical form. Users are required to provide information on “Beetle ID”, “Associated Project”, “Sampling Protocol”, “Sample Size”, “Date Collected”, and “Collected By”. Users are not required to provide any identification information with the recorded sample, although their beetle will automatically be marked as an “adult” of “unrecorded sex” if no alternative information is indicated. Users are also not required to pick a specific object as a “Source” (i.e., a tree, trap, or gallery) for their beetle. When a row is selected in the “Beetles” table, the fields in “New Beetle Record” are populated with that insect’s information (to the extent possible, not including the date collected or identified). To clear the input form, users can click the “Clear Form” button. To submit the beetle into the Review Table (where entries are held prior to final submission to the database), users should click “Submit for Review”. Beetles in the review table will accumulate with each submission until the entries are submitted or cleared.

Beetle ID *

Unique Identifier (in Scolytus or project-specific)

Associated Project *

The project the beetle is associated with

Source ID

Pick the project-associated source, if applicable

Sampling Protocol *

Select an option

Sample Size *

1

Date Collected *

May 05, 2025

Collected By *

Miranda Barnes

Vial Number

Associated Scolytus vial entry

Country

State/Province

County

Locality

Latitude

Longitude

Notes

Any other relevant information

Add Identification With Entry (Optional)

Taxon

Species name

Life Stage

Adult

Sex

Not recorded

Identification Reference

Source used to ID (optional)

Date Identified

May 05, 2025 11:13 AM

Identified By

Miranda Barnes

Notes

Any other relevant information

Make New Option

Term

New option for categorical fields

Applicable Field *

Sampling Protocol

Definition

How you want this term to be defined

Submit

Retool

Clear Form

Submit for Review

The other option for adding entries to the Review Table is to upload a file containing sample information. Users click anywhere on the “Bulk Input By File” button and are prompted to select a file from their computer. The required format is .csv. The form will accept either raw, unmapped data (i.e., categorical options are indicated by the numbers that represent them in the database) or attempt to map the information in columns based on term-value pairings. The recommendation for using this feature is to create a few dummy submissions to the review table mirroring the types of samples the user plans to bulk-enter, download either the raw or mapped data, and copy/modify it as needed in a spreadsheet software of the

user’s choice. They can then ensure that their data is properly formatted for upload.

Bulk Input By File

Select  CSV file with beetle information

Review Submission/Download Template

Beetle ID	Taxon	Life Stage	Sex	Sample Size	Sampling Protocol	Derived From	Vial #	Date Collec...	Collected ...	Project	Country	Sta
Beetle exa...	Crossotarsu...	Adult	Male	1	Field-sampled from galleries	Select option	Ente...	May 5, 2025	Mirand...	Platypodine Divers...	Indonesia	Ent

1 result

Submit AllReset Review Table

The “Review Submission/Download Template” table has columns for all entry options discussed in the “New Beetle Form”, but also includes all identification information normally shown only in the “Identifications” table (including “Date Identified”, “Identified By”, “Identification Reference”, and “Identification Notes”). All fields in the review table can be edited in the same format as in the “New Beetle Record”. Users will be prompted to save their changes as in the Projects table. **As with the Projects Table, if users do not save changes made to the Review Table, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This will also have the effect of clearing the review table. If the user wants to clear the review table without adding their entries to the database, they can click the “Reset Review Table” button.

There are several guardrails in place to make sure that users do not submit invalid data into the database through the review table – these include beetle identifiers that match material already in the database, missing information in mandatory fields (i.e., “Beetle ID”, “Sampling Protocol”, “Sample Size”, “Date Collected”, “Collected By”, or “Project”), or providing only partial information (i.e., missing “Identified By” or “Identification Date”) if you are submitting beetles with information in the “taxon” label. These “guardrails” appear in various forms in the bulk input tables on all pages and will typically direct you to make a correction based on the specific error, highlighting rows that violate database guidelines in red. You will not be able to press “Submit All” until you make appropriate changes.

Review Submission/Download Template

Beetle ID	Taxon	Life Stage	Sex	Sample Size	Sampling Protocol	Derived From	Vial #	Date Collec...	Collected ...	Proje
sample beetle 1	Xylosandrus ...	Adult	Female	1	Select option	Plant X CRASS CAMPHO...	Ente...	Feb 17, 2026	Mirand...	X. c
sample beetle 2	Xylosandrus ...	Adult	Female	1	Reared from wood frozen L...	Plant X CRASS CAMPHO...	Ente...	Feb 17, 2026	Mirand...	X. c

2 results

Please correct missing information in indicated rows under "Sampling Protocol".

Submit AllReset Review Table

USING THE PLANTS TABLE

The following tables all hold information for potential sources of the beetles entered above. They are provided in a tabbed container under the heading “Beetle Source Options” and include “Plants”, “Galleries”, and “Traps”. The “Plants” table includes columns for “Plant Label”, “Taxon”, the “Health” of the plant in question, the “Date [the plant was] Recorded”, whom the plant was “Recorded By”, the “Project”, and any “Notes” on the plant.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

Beetle Source Options

PlantsGalleriesTraps

Plant Label	Taxon	Health	Date Record...	Recorded By
Indonesia "half-alive tree"		Fallen	Oct 19, 2024	Jiri Hulcr
Indonesia dead pine		Fallen	Oct 19, 2024	Jiri Hulcr
Alabama2024SPBBark	Pinus taeda	Infested ba...	Oct 11, 2024	Miranda Barn
JIRI-XCRASS-BRANCHES		Fallen	May 20, 2024	Jiri Hulcr
OSBS SAND STUMP	Pinus clausa	Stump	May 2, 2024	Jiri Hulcr
DELUCA SAND 1	Pinus palustris	Felled	Mar 29, 2024	Jiri Hulcr
DELUCA LONGLEAF 1	Pinus palustris	Felled	Feb 29, 2024	Jiri Hulcr
OSBS LONGLEAF 1	Pinus palustris	Felled	Feb 22, 2024	Daniel Kelley
OSBS LONGLEAF 2	Pinus palustris	Felled	Feb 22, 2024	Daniel Kelley

16 results

Filtering Options

Add Plant/Tree

Plant Label *

Unique identifier

Taxon

Enter species and/or common name

Health *

Select an option

Date Recorded *

May 5, 2025

Recorded By *

Miranda Barnes

Project *

Select an option

Notes

Any other relevant information

Submit

Different plant fields can be filtered, as in Projects. The filtering options are shown below.

Filtering Options

Plant Label

Separate different search terms with semicolon-space

Taxon

Separate different search terms with semicolon-space

Health Status

Felled ✓

Fallen ✓

Stump ✓

Infested bark ✓

Date Range

Jan 1, 2019 - May 5, 2025

Lab Member

Miranda Barnes ✓

Katy Deitz ✓

Yiyi Dong ✓

Jiri Hulcr ✓

Projects

Euplatypus Project (Dummy Data) ✓

General Lab Use ✓

Platypodine Diversity ✓

SPB Fungal Symbiont Genome ✓

Notes

Separate different search terms with semicolon-space

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Recorded By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited by any logged-in user. “Taxon” and “Notes” can be edited through text input and “Health”, “Recorded By”, and “Project” can be edited using dropdown menus. “Date Recorded” can be edited through a calendar menu. All changes to the table must be saved as described above. Items can be deleted from the “Plants” table as described for the “Beetles” table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Plants” table.

CREATING A NEW PLANT

To create a new plant entry, users provide text input for “Plant Label”, “Taxon”, and “Notes” and then select from drop-down menus for “Health”, “Recorded By”, and “Project”. Plant labels must be unique. Fields can be filled automatically with information from a different tree (excepting “Date Recorded”, which will always default to today’s date) by selecting the desired row in the “Plants” table. Users must provide information for “Plant Label”, “Health”, “Date Recorded”, “Recorded By”, and “Project”. They can click submit to add the entry to the database.

USING THE GALLERIES TABLE

The “Galleries” table includes columns for “Gallery ID”, “Part Sampled”, the “Source Plant” the gallery came from, the “Date [the gallery was] Recorded”, who the gallery was “Recorded By”, the “Project”, and any “Notes” on the gallery.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

PlantsGalleriesTraps

Gallery ...	Part Sampl...	Source Plant	Date Record...	Recorded ...	Proje
GB1	Branch	JIRI-XCRASS-BRANCH...	May 24, 2024	Jiri Hulcr	X. c
GB2	Branch	JIRI-XCRASS-BRANCH...	May 24, 2024	Jiri Hulcr	X. c
GB3	Branch	JIRI-XCRASS-BRANCH...	May 24, 2024	Jiri Hulcr	X. c
1 Ic	Stump	OSBS SAND STUMP	May 2, 2024	Jiri Hulcr	Win
2 Ic	Stump	OSBS SAND STUMP	May 2, 2024	Jiri Hulcr	Win
3 Ic	Stump	OSBS SAND STUMP	May 2, 2024	Jiri Hulcr	Win
4 Ic	Stump	OSBS SAND STUMP	May 2, 2024	Jiri Hulcr	Win
5 Ic	Stump	OSBS SAND STUMP	May 2, 2024	Jiri Hulcr	Win
6 Ic	Stump	OSBS SAND STUMP	May 2, 2024	Jiri Hulcr	Win

129 results

Filtering Options

Add Gallery

Source Plant *
Select an option

Gallery ID *
Enter value

Part Sampled *
Select an option

Date Recorded *
May 5, 2025

Recorded By *
Miranda Barnes

Project *
Select an option

Notes
Any other relevant information

Submit

Different gallery fields can be filtered, as in Projects, through a dropdown menu. Filtering options are shown below.

Filtering Options

Gallery ID

Separate different search terms with semicolon-space

Date Range

Jan 1, 2019

-

May 5, 2025

Part Sampled

Trunk

Stump

Branch

Source Plant

Separate different search terms with semicolon-space

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Jiri Hulcr

Projects

General Lab Use

Platypodine Diversity

SPB Fungal Symbiont Genome

SPB Genome

Notes

Separate different search terms with semicolon-space

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Recorded By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited by any logged-in user. For entries that meet this criteria, “Part Sampled”, “Project”, and “Source Plant” can be edited using dropdown menus, while “Gallery ID” and “Notes” are edited as text and “Date Recorded” can be edited through a calendar menu. All changes to the table must be saved as described above. Items can be deleted from the “Galleries” table as described for the “Beetles” table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Galleries” table.

CREATING A NEW GALLERY

To create a new gallery entry, users provide text input for “Gallery ID”, and “Notes” and then select from dropdown menus for “Source Plant”, “Part [of the tree] Sampled”, who the gallery was “Recorded By”, and the “Project” it is associated with. Gallery labels must be unique for the given source plant they are attributed to but are not required to be universally unique. Fields can be filled automatically with information from a different gallery (excepting “Date Recorded”, which will always default to today’s date) by selecting an entry in the “Galleries” Table. Users must provide information for “Source Plant”, “Gallery ID”, “Part Sampled”, “Date Recorded”, “Recorded By”, and “Project”. Click submit to add the entry to the database.

USING THE TRAPS TABLE

The “Traps” table includes columns for “Trap ID”, “Trap Type” (ex: Lindgren), “Lure” (ex: for *Ips typographus*), “Substrate” (ex: the trap is filled with ethanol and antifreeze), “Date [the trap was] Set”, who the trap was “Set By”, the “Project” it is associated with, and any “Notes” on the trap.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in Projects. Concealed columns can be revealed by scrolling to the side.

Plants
Galleries
Traps

Trap ID	Trap Type	Lure	Substrate
OSBS-FLYING-SECOND-TRIP	Lindgren funnel tr...	Ips typographus lu...	Damp paper
DELUCA-FLYING	Lindgren funnel tr...	Ips typographus lu...	Damp paper
OSBS-FLYING	Lindgren funnel tr...	Ips typographus lu...	Damp paper

3 results

Filtering Options

Add Trap

Trap ID *

Unique identifier

Trap Type *

Select an option

Lure *

Select an option

Date Set *

May 5, 2025

Substrate *

Select an option

Project *

Select an option

Set By *

Miranda Barnes

Notes

Any other relevant information

Submit

Different “Traps” fields can be filtered, as in Projects, through a dropdown menu. Filtering options are shown below.

Filtering Options

Trap ID

Separate different search terms with semicolon-space

Date Range

Jan 1, 2019 - May 5, 2025

Trap Type

Lindgren funnel trap

Lure

Ips typographus lure

Substrate

Damp paper towel

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Jiri Hulcr

Projects

Euplatypus Project (Dummy Data)

General Lab Use

Platypodine Diversity

SPB Fungal Symbiont Genome

Notes

Separate different search terms with semicolon-space

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Recorded By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited by any logged-in user. For most recent entries, “Trap Type”, “Lure”, “Substrate”, and “Project” can be edited using drop-down menus, while “Trap ID” and “Notes” can be edited as text and “Date Set” can be altered via a calendar menu. All changes to the table must be saved as described above. Items can be deleted from the “Traps” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Traps” table.

CREATING A NEW TRAP

To create a new “Traps” entry, users provide text input for “Trap ID”, and “Notes” and then select from dropdown menus for “Trap Type”, “Lure”, “Substrate”, “Recorded By”, and “Project”. “Trap ID” entries must be unique. Fields can be filled automatically with information from a different gallery (excepting “Date

Recorded”, which will always default to today’s date) by selecting an entry in the “Traps” table. Users must provide information for all fields except for “Notes”.

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

Since so many of the dropdown menus in these forms are categorical, and all possibilities for procedures, features, etc. lab members might want to incorporate into their data cannot be preconceived by the developer, there is a “Make New Option” form nested in the Beetles form to generate new terms to use in categorical data entry options. For example, if we wanted to define a new lure, we could fill out the form with the following:

Make New Option

Term

Applicable Field *

Lure

▼

Definition

Submit

We would then click submit and observe that “New lure” now appears as an option in the Traps table under the “Lure” menu.

Add Trap

Trap ID *

Unique identifier

Trap Type *

Select an option

Date Set *

May 5, 2025

Project *

Select an option

Notes

Any other relevant information

Lure *

Select an option

lps typographus lure

new lure

Set By *

Miranda Barnes

Submit

We also see that the definition for the new categorical term appears at the bottom of the page in “Table Terms”.

Table Terms	
Term	Definition
Reared from wood into ethanol	Beetles were collected live from wood removed from the site and placed in ethanol for long-term storage
Sex Unknown	The sex of the sampled beetle(s) are unknown
Stump	Dead tree, cut, only stump remaining (as a plant health status indicator)
Stump	The stump of a felled tree (as a part from which galleries are sampled)
Trunk	Sampled from the stem of a fallen/felled tree
new lure	This is a fake lure that should be deleted later
30 results	

New categorical options can be defined for “Sampling Protocol” (“Beetles”), “Life Stage” (“Beetles”), “Sex” (“Beetles”), “Health” (“Plants”), “Trap Type” (“Traps”), “Lure” (“Traps”), “Substrate” (“Traps”), and “Part Sampled” (“Galleries”). This table can be downloaded as-is and filtered using the options below:

Filtering Options

Term

Definition

Separate different search terms with semicolon-space

Separate different search terms with semicolon-space

Note that **unless you immediately use the terms you define in fields** in beetle/tree/gallery/trap input, **you will not be able to access any “New Options” you’ve created when you refresh the page**. The goal here is to avoid accumulating purely hypothetical or “junk” terms in input forms that are not actually tied to objects in the database. However, as soon as you submit a sample with a new option you define, it will be available as long as that sample is in the database.

MEDIA PREP PAGE

The “Media Prep” page stores information on media that are available for use in cultures and slants. This page exists for a few reasons. Firstly, it is meant to enable tracking of batches of plates to ensure that appropriate leftover media can be shared among lab members and that old media can be better identified and discarded. Secondly, it permits some degree of quality control by permitting users to track whether plates with similar contamination issues might come from the same batch. Lastly (and most importantly), it encourages lab members to be assiduous in recording the composition of the media they’re using, which can be important when determining what fungi grow and how well. Even something as simple as malt extract agar (MEA) can vary between publications and lab members based on the relative ratios of malt extract to agar to volume of water used. These slight disparities may be unlikely to make a large difference in fungal growth, but in principle, media composition should be consistent between experiments and recorded beyond vague abbreviations.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE MEDIA PREP TABLE

The “Media Prep” table includes fields for “Prep ID” (displaying the unique identifier assigned to the media batch), the “Type” (the recognizable abbreviation associated with a general media category, like MEA or PDA), the “Vessel” (i.e., whether it was poured into a large plate, a small plate, a slant, etc.), the “Pre-Autoclave Ingredients” (i.e., materials like malt extract and agar added prior to autoclaving), “Post-Autoclave Ingredients” indicating if antibiotics or vitamins were added to the medium, “Quantity” (reflecting how many plates were made), the person the medium was “Prepared By”, the “Project” the medium was originally prepared for, the “Date” it was prepared on, and any additional “Notes” associated with the medium.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in Projects. Concealed columns can be revealed by scrolling to the

side.

Media Prep Dashboard				
Prep ID	Type	Vessel	Pre-Autoclave Ingredients	Post-Autoclave Ingrec
39	PDA	large plate	2 x (23.4 g Disco PDA in 600 mL)	2 x (.06 g streptomycin)
35	Cyc	large plate	23 g Disco PDA in 600 mL water	.06 g streptomycin sul
36	PDA	large plate	39 g Disco PDA in 600 mL water	
34	PDA	large plate	23 g Disco PDA in 600 mL water	
37	PDA	small plate	23 g Disco PDA in 600 mL	
32	MYEA	large plate	12 g agar, 12 g malt extract, 1.2 g yeast extract in 600 mL water	
22	MEA	large plate	12 g agar, 12 g malt extract	
20 results				

Different “Media Prep” fields can be filtered, as in Projects, through a dropdown menu. Filtering options are shown below.

Filtering Options

Prep ID

Separate different search terms with semicolon

Type

MYEA

PDA

dH2O

BSMA

MEA

Vessel

large plate

cryogenic vial -...

cryogenic vial -...

small plate

Erlenmeyer flask

Quantity

1-150

Date Range

Jan 1, 2019 - May 7, 2025

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Belinda

Projects

General Lab Use

Platypodine Di...

SPB Fungal Sy...

Pre-Autoclave Ingredients

Separate terms with semicolon-space

Post-Autoclave Ingredients

Separate terms with semicolon-space

Notes

Separate terms with semicolon-space

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Prepared By” column is the same as the one who is currently logged in). The only exceptions to this rule are “Quantity” and “Notes”, which can be edited by any logged-in user. “Pre-Autoclave Ingredients”, “Post-Autoclave Ingredients”, and “Notes” can be edited as text. “Type”, “Vessel”, “Prepared By”, and “Project” are edited through a dropdown menu. “Date Set” can be altered via a calendar menu. All changes to the table must be saved as described above. Items can be deleted from the “Media Prep” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Media Prep” table.

CREATING A NEW MEDIA BATCH

To create a new “Media Batch” entry, users provide text input for “Pre-Autoclave [Ingredients]”, “Post-Autoclave [Ingredients]”, and “Notes” and then select from drop-down menus for “Type”, “Vessel”, “Prepared By”, and “Project”. “Quantity” takes numerical input. “Prep ID” is generated automatically. Fields can be filled automatically with information from a preceding Media Batch (excepting “Date Recorded”, which will always default to today’s date) by selecting an entry in the “Media Batch” table. Users must provide information for all fields except for “Notes” and “Post-Autoclave” (since antibiotics and the like are not always added to media). Users click “Submit” to add the new entry to the database.

New Batch

Type *

Vessel *

Starting Quantity *

Select media type

Select media vessel/format

1

Pre-Autoclave *

Exact composition prior to autoclaving

Post-Autoclave

Materials added after autoclaving (i.e., antibiotics)

Project *

Prepared By *

Date Prepared *

Select an option

Miranda Barnes

May 7, 2025

Notes

Enter value

Submit

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

As with “Beetles”, additional options for categorical entry fields “Type” and “Vessel” can be defined in “Make New Option” at the bottom of the page. Definitions for previously created options are also shown in “Table Terms” at the bottom of the page. This table can be filtered and downloaded as-is.

CULTURES PAGE

The Cultures page stores information on short-term (primarily fungal) cultures created by lab members. This contrasts with the Slants page, which stores information on long-term cultures in the freezer, etc. “Cultures” also contains information on morphotypes identified on different plates.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE CULTURES TABLE

The “Cultures” table includes fields for “Physical Label” (the text written on the plate to identify it), “Media” (the media batch that the plate came from), the “Source” used as inoculum for the plate, the “Tissue” that was used to inoculate the plate (i.e., for example, the mycangium of a beetle, the whole beetle, mycelium from another plate, etc.), the “Technique” that was used to inoculate the plate (i.e., dilution, streak plate, etc.), who the plate was “Prepared By”, the “Project” it was prepared for, the “Incubation Temperature”, the

“Date [the plate was] Created”, the “Fridge Date” describing if or when the plate was moved from the incubator into the fridge for longer-term storage, the “Discard Date” describing if and when the plate was disposed of, and any additional “Notes” associated with the culture.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in Projects. Concealed columns can be revealed by scrolling to the side.

Cultures Dashboard							
Physical Label	Media	Source	Tissue	Technique	Prepared By	Project	
P999 PDA from INDO-3-PRONOTUM, .01, 2025-04-...	36: PDA large pla...	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	prothor...	dilution-.01	Katy Dei...	Platyp...	
P998 PDA from INDO-3-PRONOTUM, .1, 2025-04-17	36: PDA large pla...	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	prothor...	dilution-.1	Katy Dei...	Platyp...	
P997 Cyc from INDO-3-HEAD, .001, 2025-04-17	35: Cyc large plate	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	head	dilution-.0...	Katy Dei...	Platyp...	
P996 Cyc from INDO-3-HEAD, .01, 2025-04-17	35: Cyc large plate	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	head	dilution-.01	Katy Dei...	Platyp...	
P995 Cyc from INDO-3-HEAD, .1, 2025-04-17	35: Cyc large plate	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	head	dilution-.1	Katy Dei...	Platyp...	
P994 PDA from INDO-3-HEAD, .001, 2025-04-17	36: PDA large pla...	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	head	dilution-.0...	Katy Dei...	Platyp...	
P993 PDA from INDO-3-HEAD, .01, 2025-04-17	36: PDA large pla...	Beetle INDO, Mt. Salak 2025-04-15 Hulcr & Narakusumo ...	head	dilution-.01	Katy Dei...	Platyp...	
1,860 results							

Different culture fields can be filtered, as in “Projects”. Filtering options are shown below. Note that when start and end dates are applied to the “Fridge Date” and “Discard Date”, cultures without values in this column (i.e., cultures still in the incubator) will be completely excluded until you clear both of these values again.

Filtering Options

Physical Label

Separate different search terms with semicolon-space

Media

1: MYEA large plate

2: MYEA large plate

Show entries with empty media fields.

Source

Separate different search terms with semicolon-space

Tissue

mycangium

edge of culture

whole insect

middle of culture

Technique

dilution-.1

dilution-.01

dilution-.001

loop

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Jiri Hulcr

Projects

Ambrosiella Phase Transition

Cycloheximide Concentration

Dinoplatypus Project

Euplatypus Project (Dummy Data)

Notes

Separate different search terms with semicolon-space

Incubation Temperature

24-25

Date Range (Date Created)

Mar 24, 2006 - Feb 17, 2026

Date Range (Fridge Date)

Start Date - End Date

Date Range (Discard Date)

Start Date - End Date

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Prepared By” column is the same as the one who is currently logged in). The exceptions to this rule are “Fridge Date”, “Discard Date”, and “Notes”, which can be edited by anyone logged in. “Physical Label” and “Notes” can be edited as text, “Incubation Temperature” can be edited numerically, and “Media”, “Source”, “Tissue”, “Technique”, “Prepared By”, and “Project” can be edited through a dropdown menu. “Date Created”, “Fridge Date”, and “Discard Date” are modified through a calendar menu. All changes to the table must be saved as described above. Items can be deleted from “Cultures” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

However, it is worth noting that in addition to delete, upon selecting a plate, two additional options appear: “Put Plate(s) in Fridge” and “Discard Plate(s)”. These are shortcuts for moving multiple plates into the fridge or physically discarding plates (but not deleting the plate entry in the database). These buttons will automatically set the fridge and discard dates to whatever is specified in the “Selected Bulk Fridge/Discard Date” below the buttons (defaulting to today’s date). However, if users need to clear fridge or discard date from a plate that was not actually put in the fridge or discarded, they need to edit samples individually in the table.

Cultures Dashboard

Physical Label	Media	Source	Tissue	Technique	Project
P1833 PDA 0 from S17756 Ophiostoma minus, 2026-02-16	68: PDA large pla...	Slant 17756	frozen ...	chunk	
P1832 MEA pH 9 from P1768 FM, 2026-02-16	77: MEA 9 small ...	Culture P1768 PDA 0 from S18489 Aroep, 2026-02-09	edge of...	chunk	
P1831 MEA pH 9 from P1768 FM, 2026-02-16	77: MEA 9 small ...	Culture P1768 PDA 0 from S18489 Aroep, 2026-02-09	edge of...	chunk	
P1830 MEA pH 9 from P1768 FM, 2026-02-16	77: MEA 9 small ...	Culture P1768 PDA 0 from S18489 Aroep, 2026-02-09	edge of...	chunk	
P1829 MEA pH 9 from P1768 FM, 2026-02-16	77: MEA 9 small ...	Culture P1768 PDA 0 from S18489 Aroep, 2026-02-09	edge of...	chunk	
P1828 MEA pH 9 from P1768 FM, 2026-02-16	77: MEA 9 small ...	Culture P1768 PDA 0 from S18489 Aroep, 2026-02-09	edge of...	chunk	
P1827 MEA pH 9 from P1768 FM, 2026-02-16	77: MEA 9 small ...	Culture P1768 PDA 0 from S18489 Aroep, 2026-02-09	edge of...	chunk	
1 of 1,398 selected					
Delete Plate(s)		Put Plate(s) in Fridge		Discard Plate(s)	
Selected Bulk Fridge/Discard Date:			Feb 17, 2026		

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Cultures” table.

USING THE MORPHOTYPES TABLE

The Morphotypes table sits directly under the “Cultures” table in the dropdown “Morphotypes” container and holds information about morphotypes assigned to colonies on different plates. Morphotypes for particular plates can be viewed by clicking the checkbox beside “Physical Label” on the Cultures Table. Entries include fields for a rough physical descriptor of the “Morphotype”, the physical label on the “Original Plate” that the morphotype was recorded from, the number of colony-forming units (“CFUs”) present on that plate, the “Date Observed”, the person it was “Recorded By”, and the “Project” it was recorded as a part of, as well as any “Notes” that a user might have on a morphotype.

Morphotypes

Morphotype	Original Plate	CFUs	Date Observed	Recorded By
M6 - Euplatypus ...	21376	4,9...	Jun 18, 2022	Andre Rodi
M2 - Euplatypus ...	21376	1,5...	Jun 18, 2022	Andre Rodi
M3 - Euplatypus ...	21376	10	Jun 18, 2022	Andre Rodi
M5 - Euplatypus ...	21378	60	Jun 18, 2022	Andre Rodi
M3 - Euplatypus ...	21378	170	Jun 18, 2022	Andre Rodi
M1 - Euplatypus ...	21378	1,4...	Jun 18, 2022	Andre Rodi
M3 - Euplatypus ...	21380	2,6...	Jun 18, 2022	Andre Rodi
M1 - Euplatypus ...	21380	40	Jun 18, 2022	Andre Rodi
M6 - Euplatypus ...	21380	10	Jun 18, 2022	Andre Rodi
M4 - Euplatypus ...	21380	70	Jun 18, 2022	Andre Rodi
M2 - Euplatypus ...	21383	26,...	Jun 18, 2022	Andre Rodi

170 results

Add Morphotype to Plate

Chosen Plate *

21663

Morphotype *

Pick existing morphotype or create new one via the new options form

CFUs *

1

Notes

Any additional information

Date Observed *

May 10, 2025

Submit

Different fields can be filtered, as in Projects. Filtering options are shown below.

Filtering Options

Morphotype Options

M1 - Euplatypus Project (2019)

✓

M2 - Euplatypus Project (2019)

✓

M3 - Euplatypus Project (2019)

✓

M4 - Euplatypus Project (2019)

✓

M5 - Euplatypus Project (2019)

✓

M6 - Euplatypus Project (2019)

✓

M7 - Euplatypus Project (2019)

✓

Date Range

Jan 1, 2019 - May 11, 2025

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Projects

Euplatypus Project (Dummy Data)

✓

General Lab Use

Platypodine Diversity

Notes

Separate terms with semicolon-space

Sample Size

1-67,000

There are a couple of options for creating and modifying morphotypes. The first option is to add or edit morphotypes individually. To create a new individual morphotype (with notes), you can use the “Add Morphotype to Plate” form and select from a list of available plates (limited to the ones currently selected and defaulting to the last selected row), another dropdown menu of available morphotypes, indicate the number of “CFUs” (numerical input), any additional “Notes” as text, and then indicate the “Date Observed” (defaulting to today’s date) in a calendar menu before pressing submit. The project defaults to the one associated with the original plate, and the lab member defaults to the current user.

To edit an individual morphotype, you can edit the basic morphotypes table. As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Recorded By” column is the same as the one who is currently logged in). The one exception to this rule is “Notes”, which can be edited by anyone logged in. “Notes” can be edited as text, “CFUs” can be edited numerically, and “Morphotype”, “Recorded By”, and “Project” can be edited through a dropdown menu. “Date Observed” is modified through a calendar menu. “Original Plate” cannot be edited. All changes to the table must be saved as described above. Items can be deleted from “Morphotypes” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

Alternatively, you can add, edit, and delete morphotypes in the “Morphotypes Bulk Editor”. This displays as a table where the plates are rows and the morphotypes recorded on each one are columns; in each of the cells, the number of a particular morphotype per plate is displayed. When you edit the numbers in this table to reflect and click the save prompt, one of three things will happen. If you indicated that a new non-zero number of CFUs on a plate, then it will create a new morphotype entry for that plate with the corresponding CFU number. If you change the number of CFUs on a plate for a morphotype you’ve already recorded CFUs for, it will update the number of CFUs in that morphotype entry accordingly. If you set a CFU value that was previously a non-zero number to zero, that morphotype will be deleted from the database. To adjust the morphotypes shown in the table, select and deselect entries in the “Morphotype Options” form.

Morphotypes Bulk Editor

Physical Label	M3 - Euplaty...	M2 - Euplaty...	M6 - Euplaty...	M1 - Euplaty...	M5 - Euplaty...
21695	0	0	0	0	0
21656	0	0	0	0	0
21666	0	0	0	0	0
21376	10	1510	4920	0	0
21421	0	0	0	0	0
21428	200	0	0	40	0
376 results					

Morphotype Options

M1 - Euplatypus Project (2019)	✓
M2 - Euplatypus Project (2019)	✓
M3 - Euplatypus Project (2019)	✓
M4 - Euplatypus Project (2019)	✓
M5 - Euplatypus Project (2019)	✓
M6 - Euplatypus Project (2019)	✓
M7 - Euplatypus Project (2019)	✓
M8 - Euplatypus Project (2019)	✓

CREATING A NEW CULTURE

There are two ways to add new “Cultures” to the database: singly or in bulk. To draft individual or “clonal” entries, there is a “Culture New Plate” form. Users provide text input for “Physical Label” and “Notes” and select from a dropdown menu for “Associated Project”, “Media Prep Batch” (only available media are displayed), “Tissue”, “Technique”, “Source ID”, and “Prepared By”. They are expected to alter the date in “Date Inoculated” if it is different from the default (today’s date) and to provide numerical input for “Incubation Temperature” (in Celsius). Lastly, they should select the “Number of Plates” by typing or using the stepper at the bottom of the form – for example, if a person plans to generate many subcultures that are identical to one another, they can specify the quantity of subcultures they’re preparing in that field. When the user picks an “Associated Project”, the options in the “Source ID” field change to include only objects derived from that project.

If a user wants to create a plate for a new project using a source from an old project, they should select the source of the old project and then edit it to reflect the desired old project in the “Review Submission/Download Template Table”. Alternatively, they can leave “Source ID” empty and edit using the dropdown list containing all possible sources once it is submitted to the review table.

“Physical Label” automatically provides a suggestion for what information users should write on the plate itself as the form is completed, and it also provides a sequential number to associated with that plate. The user can either edit this label in the form after all other fields have been specified or, alternatively, submit it to the review table and edit it there (this is the only way to remove the automatically generated plate number from the physical label unless you are using the bulk submitter). Users are required to provide information on “Associated Project”, “Media Prep Batch”, “Tissue”, “Technique”, “Incubation Temperature”, “Date Inoculated”, “Prepared By”, and “Physical Label”. “Number of Plates” will default to 1. To automatically fill the table with information from a previous plate, users should click the checkbox beside that plate in the “Cultures” table.

To submit the culture into the Review Table (where entries are held prior to final submission to the database), users should click “Submit for Review”. Users can continue to add cultures to the review table by repeatedly submitting this form with different information – cultures in the review table will accumulate with each submission until the “Review Submission/Download Template” Table is submitted or cleared. To clear the input form, users can click the “Clear Form” button (this will not clear information in the Review Table).

Culture New Plate

Associated Project *

The project the plate is associated with

Media Prep Batch *

Media being used

Tissue *

hindgut

Technique *

roll

Source ID

Beetle 37774

Incubation Temperature *

25

°C

Date Inoculated *

May 10, 2025

Prepared By *

Miranda Barnes

Physical Label *

P1142 from B37774, 2025-05-10

Notes

Any other relevant information

Number of Plates

1

Clear Form

Submit for Review

Make New Option

Term

New option for categorical fields

Definition

How you want this term to be defined

Applicable Field *

Select field

Submit

Bulk Input By File

Select

CSV file with plate information

The other option for adding entries to the Review Table is to use the “Bulk Input by File” option. Users click anywhere on the “Bulk Input By File” button and are prompted to select a file from their computer. The required format is .csv. The form will accept either raw, unmapped data (i.e., categorical options are indicated by the numbers that represent them in the database) or will attempt to map the information in columns according to value-term pairs the database recognizes. The recommendation for using this feature is to create a few dummy submissions to the review table mirroring the types of samples the user plans to bulk-enter, then download either the raw or mapped data and copy/modify it as needed in a spreadsheet software of the user’s choice. They can then ensure that their data is properly formatted for upload.

The “Review Submission/Download Template Table” has columns for all entry options discussed in the “Culture New Plate” form. All fields in the Review Table can be edited in the same format as in “Culture New Plate”. Users will be prompted to save their changes as in the Projects table. **As with the Projects Table, if users do not save changes made to the Review Table, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

Review Submission/Download Template								
Physical Label	Media	Source	Tissue	Technique	Date Created	Prepared By	Project	Incubation Temperature
P1142 BSMA from B37774, 2025-05-10	12: BSMA...	Beetle 37774	hindgut	roll	May 10, 2025	Miranda B...	General L...	25

1 result

Submit All Reset Review Table

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This will also have the effect of clearing the Review Table. If the user wants to clear the review table without adding their entries to the database, they can click the “Reset Review Table” button.

As with the “Beetles” Review Submission table, if a duplicate “Physical Label” or there is missing information from required fields (i.e., “Associated Project”, “Media Prep Batch”, “Tissue”, “Technique”, “Incubation Temperature”, “Date Inoculated”, “Prepared By”, and “Physical Label”), you will not be able to “Submit All” and problematic rows will be highlighted in red until you make and save appropriate edits.

Review Submission/Download Template								
Physical Label	Media	Source	Tissue	Technique	Date Created	Prepared By	Project	Incubation Temperature
P1834 MEA 9 from CP1768, 2026-02-17	77: MEA ...	Culture P1768 PDA 0 from S18489 Aroep, 2026...	edge of c...	chunk	Feb 17, 2026	Miranda B...	Ambr	
P1835 MEA 9 from CP1768, 2026-02-17	77: MEA ...	Culture P1768 PDA 0 from S18489 Aroep, 2026...	Select option	chunk	Feb 17, 2026	Miranda B...	Ambr	

2 results

Please correct missing information in indicated rows under "Tissue".

Submit All Reset Review Table

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

As with the “Beetles” page, additional options for categorical entry fields “Technique” (“Cultures”), “Tissue” (“Cultures”), and “Morphotype” (“Morphotypes”) can be defined in “Make New Option” between the “Add Morphotype to Plate” form and the “Review Submission/Download Template” table. Definitions for previously created options are also shown in “Table Terms” at the bottom of the page. This table can be filtered and downloaded as-is.

SLANTS PAGE

The “Slants” page stores information on various long-term cultures kept as voucher specimens or as backup for certain types of experiments. It also includes a short form for culturing a plate from a slant, though the

results of this action will not appear on the “Slants” page and will require the user to go to the “Cultures” page to confirm what their submission looks like and edit it accordingly.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE SLANTS TABLE

The “Slants” table includes fields for “Slant ID” (displaying the text or number actually written on the slant), “Media” (the media batch that the slant was inoculated on), the “Source” plate used as inoculum, the “Location” of the slant (for example, in the incubator, moved to the freezer, etc.), the “Storage Method” used to preserve the culture, the “Date [the slant was] Created”, who the plate was “Prepared By”, the “Project” it was prepared for, and any additional “Notes” associated with the slant.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

Slants Dashboard				
Slant ID	Media	Source	Location	Storage
IND-4	15: MEA large plate	P321 Cyc from 42513.1 ...	Incubator	Long-t
IND-5	15: MEA large plate	P352 Cyc from 42513.0...	Incubator	Long-t
Aro1024	21: PDA cryogenic vial - slant	P207 PDA CP from S184...	Fridge	Long-t
Cer1024	21: PDA cryogenic vial - slant	P188 PDA from S18477,...	Freezer (-80 degrees Celsi...	Frozen
18479	6: PDA cryogenic vial - slant	MYEA w/strep CP Ent ba...	Freezer (-80 degrees Celsi...	Frozen
18480	6: PDA cryogenic vial - slant	P33 PDA w/strep tall fuz...	Freezer (-80 degrees Celsi...	Frozen
18481	6: PDA cryogenic vial - slant	P34 PDA w/strep raised ...	Freezer (-80 degrees Celsi...	Frozen
18482	6: PDA cryogenic vial - slant	P38 PDA w/strep 21896 ...	Freezer (-80 degrees Celsi...	Frozen
74 results				 

Different “Slants” fields can be filtered, as in Projects, through a dropdown menu. Filtering options are shown below.

Filtering Options

Slant ID

Separate different search terms with semicolon-space

Date Range

Jan 1, 2019 - May 8, 2025

Media

1: MYEA large plate

2: MYEA large plate

3: PDA large plate

4: MYEA large plate

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Jiri Hulcr

Projects

SPB Mycangial Transcriptome

Wingfield Sabbatical

X. crass Genome

X. crassiusculus Fungal Symbiont Ge...

Source

Separate different search terms with semicolon-space

Notes

Separate different search terms with semicolon-space

Location

Discarded

Freezer (-80...

Fridge

Incubator

Storage Method

Frozen in glycerol slant

Long-term refrigeration

Water at room temperatu...

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Prepared By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited as text by any logged-in user. “Media”, “Source”, “Storage Method”, “Location”, “Prepared By”, and “Project” can be edited through a dropdown menu. “Slant ID” can be edited as text, and “Date Created” can be altered via a calendar menu. All changes to the table must be saved as described above. Items can be deleted from the “Slants” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Slants” table.

CREATING A NEW SLANT

There are two ways to add new “Slants” to the database: singly or in bulk. To draft individual entries, there is a “New Slant” form. Users provide text input for “Slant ID” and “Notes” and select from dropdown menus for “Associated Project”, “Media Prep Batch” (only available media are displayed), “Source ID”, “Location”, “Storage Method”, and “Prepared By”. They are expected to alter the date in “Date Created” if it is different from the default of today’s date. When the user picks an “Associated Project”, the “Source ID” field is populated with only sample information generated for that project. Users are required to provide information on “Associated Project”, “Location”, “Storage Method”, “Prepared By”, “Date Created”, and “Slant ID”. Users do not have to specify a “Media Prep Batch” or “Source ID”. This is intended to better accommodate slants from projects preceding the introduction of the database. One can populate the form with information from an existing slant by clicking on the checkbox beside that entry on the left-hand side of the “Slants” table.

New Slant

Associated Project *	The project the "slant" is associated with
Media Prep Batch	Media being used
Source ID	Where the slant was derived from
Location *	Where the material is being kept
Storage Method *	The method used to store the material
Date Created *	May 08, 2025
Prepared By *	Miranda Barnes
Notes	Any other relevant information
Slant ID *	How the slant is identified (number/label)

Clear Form

Submit to Review Table

Bulk Input By File

Select

CSV file with slant information

To submit the slant into the Review Table (where entries are held prior to final submission to the database), users should click “Submit for Review”. Users can continue to add slants to the review table by repeatedly submitting this form with different information – slants in the review table will accumulate with each submission until the “Review Submission/Download Template” Table is submitted or cleared. To clear the input form, users can click the “Clear Form” button (this will not clear information in the Review Table).

The other option for adding entries to the Review Table is to use the “Bulk Input by File” option. This works in the same way that the “Beetles”, “Cultures” and “Slants” bulk submit features do.

The “Review Submission/Download Template Table” has columns for all entry options discussed in the “New Slant” form. All fields in the Review Table can be edited in the same format as in this form. Users will be prompted to save their changes as in the Projects table. **As with the Projects Table, if users do not save changes made to the Review Table, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

Slant ID	Media	Source	Location	Storage Met...	Date Creat...	Prepared By	Project
example slant	29: MYEA...	Culture P449 M...	Freezer (-80 degrees Celsius)	Frozen in ...	May 8, 2025	Miranda Barnes	Platypodine Diversity

1 result

Submit All Reset Review Table

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This will also have the effect of clearing the Review Table. If the user wants to clear the review table without adding their entries to the database, they can click the “Reset Review Table” button.

As with preceding pages, if there are duplicate “Slant IDs”, or there is missing information from required fields (i.e., “Project”, “Location”, “Storage Method”, “Prepared By”, “Date Created”, and “Slant ID”), you will not be able to “Submit All.” Problematic rows will be highlighted in red until you make and save appropriate edits.

Review Submission/Download Template								
Slant ID	Media	Source	Location	Storage Met...	Date Creat...	Prepared By	Project	Note
sample slant	Select option	Select option	Freezer (-80 degrees Celsius)	Frozen in ...	Feb 17, 20...	Miranda Barnes	Ambrosiella Phase Transition	Ente
sample slant 2	Select option	Select option	Freezer (-80 degrees Celsius)	Select option	Feb 17, 20...	Miranda Barnes	Ambrosiella Phase Transition	Ente

2 results

Please correct missing information in indicated rows under “Storage Method”.

Submit All Reset Review Table

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

As with the “Beetles” page, additional options for categorical entry fields “Storage Option” and “Location” can be defined in the “Make New Option” form between the “Slants” table and the “Culture New Plate from Slant” form. Definitions for previously created options are also shown in “Table Terms” at the bottom of the page. This table can be filtered and downloaded as-is.

CULTURE NEW PLATE FROM SLANT

Because samples may be “resurrected” from slants into short-term culture, an option to create a plate from entries on the “Slants” page has been made available. Users can either select a slant from the drop-down

menu or a row within the “Slants” table to populate this form with information. Users also have drop-down menus to select information for “Project”, “Media Batch”, and “Prepared By”, with numerical input for “Temperature” (i.e., incubation temperature in Celsius) and date selection for “Date”. To create the new plate, users click “Submit” at the bottom of the form.

EXTRACTIONS PAGE

The “Extractions” page stores information on DNA and RNA extractions from beetles, galleries, cultures, etc. Extractions are uniquely identified by a combination of “Batch Label” and “Tube #”, and, unlike the entries addressed so far, are explicitly designed to accept multiple sample sources.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE EXTRACTIONS TABLE

The “Extractions” table includes fields for “Batch Label” (general name for a particular set of extractions, ideally conducted on the same day using the same method), the “Tube #” (specifying which extraction is which within a given batch – the assumption is that this will be numeric, but the database also accepts text input), the “Source” (or sources) of tissue for extraction, the “Method” used to generate the extraction (i.e., Extract-N-Amp, CTAB, etc. – preferably in as much detail as possible), the “Date [the extraction was] Started” (since some extractions may take several days to complete), whom the plate was “Prepared By”, the “Project” it was prepared for, and any additional “Notes” associated with the extraction.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

Extractions Dashboard					
Batch Label	Tube #	Source	Method	Date Started	Prepared By
03/04/2025 summer school	1	Culture P492 MYEA from P459 Geosmithia Ma, 2025-02-08	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
03/04/2025 summer school	2	Culture P511 MYEA from P455, Ambrosiozyma sp., Ex, 2025-02-18	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
03/04/2025 summer school	3	Culture P504 MYEA from P449, Neocosmospora ambrosium, Ma, 2025-0...	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
03/04/2025 summer school	4	Culture P525 MYEA from P515, Irpex ambrosius Ex, 2025-02-26	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
03/04/2025 summer school	5	Culture P503 MYEA from P462 Ex E. cobbii, 2025-02-14	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
03/04/2025 summer school	6	Culture P508 MYEA from P448, Raffaelea xyleborini, Ex, 2025-02-18	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
03/04/2025 summer school	7	Culture P509 MYEA from P448, Raffaelea xyleborini, Ph, 2025-02-18	Extract-n-Amp with BSA	Mar 5, 2025	Miranda Bi
1/20/25 CTAB DNA	1-114-2	Culture P511 MYEA from P455, Ambrosiozyma sp., Ex, 2025-02-18	CTAB DNA extraction	Jan 20, 2025	Miranda Bi
238 results					ⓘ ⚙

Different “Extractions” fields can be filtered, as in “Projects”. Filtering options are shown below.

Filtering Options

Batch Label

Separate different search terms with semicolon-space

Tube #

Lab Member

Miranda Barnes

Katy Deitz

Yiyi Dong

Jiri Hulcr

Andrew Johnson

Projects

Euplatypus Project (Dummy Data)

General Lab Use

Platypodine Diversity

SPB Fungal Symbiont Genome

SPB Genome

Source

Separate different search terms with semicolon-space

Method

CTAB RNA extraction

CTAB for HMW DNA

E.Z.N.A.® Fungal RNA Mini Kit

E.Z.N.A.® Mollusc & Insect DNA Kit

Date Range

Jan 1, 2019 - May 11, 2025

Notes

Separate different search terms with semicolon-space

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Prepared By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited as text by any logged-in user. “Batch Label”, “Tube #”, “Source”, “Method”, “Prepared By”, and “Project” can be edited through a dropdown menu, and “Date Started” can be edited with a calendar menu. Multiple options in “Source” may be selected and/or removed at once. Users must click “Save” for their changes to be preserved. Extractions can be deleted from the “Extractions” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Extractions” table.

CREATING NEW EXTRACTIONS

Because extractions are typically conducted in bulk – i.e., the details of the method are typically the same, but the source samples may be different - the “Extractions” sample creation framework relies heavily on an “Input Table” to view and edit prospective entries.

There are two options for filling the input table. The first is to use the “Create New Batch” form. When using this system, users select a quantity for “Number of Tubes” (defaulting to 8 as the standard strip tube length) and provide text input for “Batch Label” and “Notes”. They use dropdown menus to specify the “Method” they are using, the associated “Project”, and who it was “Prepared By”. (They can also modify the “Date Started” accordingly). When they click “Generate Batch”, the “Input Table” will populate with the information they have just provided. Any of the fields (including “Tube Number”) can be edited, but all source fields **must**

be individually specified using a dropdown menu and saved to the table before submission.

Create New Batch

Number of Tubes

8

Batch Label

null

Method

The extraction protocol used

Project

Select an option

Date Started

Aug 12, 2024

Prepared By

Select an option

Notes

Any additional information

Generate Batch

New Method

Term

New option for categorical fields

Definition

How you want this term to be defined

Submit

Bulk Input By File

Select

CSV file with extractions information

Review Submission/Download Template								
Batch Label	Tube Number	Method	Source		Date Started	Prepared By	Project	Notes
example	1	Extract-n-Amp with BSA	Culture P342 KMD-IND-4	Culture P343 KMD-IND-5	May 11, 2025	Miranda B...	X. crassiu...	
example	2	Extract-n-Amp with BSA	Culture DL-TR6		May 11, 2025	Miranda B...	X. crassiu...	
example	3	Extract-n-Amp with BSA	Culture DL-TR16		May 11, 2025	Miranda B...	X. crassiu...	
3 results								
Submit AllReset Input Table								

The second option is to upload a file containing sample information. Users click anywhere on the “Bulk Input By File” button and are prompted to select a file from their computer. The required format is .csv. The form will accept either raw, unmapped data (i.e., categorical options are indicated by the numbers the database uses to represent them) or will attempt to map the information in columns according to terms it recognizes. The recommendation for using this feature is to create a few dummy submissions in the review table mirroring the types of samples the user plans to bulk-enter, download either the raw or mapped data, and then copy/modify it as needed in a spreadsheet software of the user’s choice. You can then ensure that your data is properly formatted for upload.

Users will be prompted to save any changes made, as in the “Projects” table. **As with previous bulk submit tables, if users do not save changes made to the “Review Submission/Download Template”, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This also has the effect of clearing the “Input Table”. If the user wants to clear the “Input Table” without adding their entries to the database, they can click the “Reset Input Table” button.

As with preceding pages, if a duplicate “Batch Label” + “Tube Number” combination exists or there is missing information from required fields (i.e., “Batch Label”, “Tube Number”, “Method”, “Source”, “Date Started”, and “Project”), you will not be able to “Submit All” and problematic rows will be highlighted in red until you make and save appropriate edits.

Review Submission/Download Template

Batch Label	Tube Number	Method	Source	Date Started	Prepared By	Project	Notes
sample batch	1	Monarch PCR & DNA c...	Beetle MGB20250204R	Feb 18, 2026	Miranda B...	General L...	
sample batch	2	Monarch PCR & DNA c...	Beetle MGB20250204Q	Feb 18, 2026	Miranda B...	General L...	
sample batch	3	Monarch PCR & DNA c...	Select options	Feb 18, 2026	Miranda B...	General L...	

3 results

Please correct missing information in indicated rows under "Source".

Submit All
Reset Input Table

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

On the “Extractions” page, there is only one categorical field to which users are permitted to add new options: “Method”. Hence, the “New Options” form is reduced to a “New Method” form. It is otherwise operated identically. Generated terms and definitions are displayed in “Table Terms” at the bottom of the page.

Because it may not be convenient to specify a method in as much detail as would be preferred in the term and definition fields, users are encouraged to provide links to documents or webpages containing more extensive information rather than only providing vague descriptions of what they mean by a given procedure.

PCR PRODUCTS FROM EXTRACTATIONS

When you click to select entries in the central “Extractions” table, you’ll also see a form pop up beneath “Filtering Options” and beside “Create New Batch” offering to help the user create “PCR From Selected Extractions”. This is to speed input on the subsequent “PCR Products” page. Users select a project, forward primer, reverse primer, and cycle and provide a batch label and the name of the person preparing the

sample, as well as any additional notes they may have on the extractions.

PCR From Selected Extractions

Project *

Select an option

Batch Label *

Label associated with whole PCR batch ir

Forward Primer *

Select an option

Reverse Primer *

Select an option

Cycle *

Select an option

Prepared By *

Select an option

Notes

Enter value

Download Table

When a user presses “Download Table”, they receive a .csv file that can be brought to the subsequent “PCR Products” page as a template for the submission table there. Tube numbers are automatically carried over and must be edited manually if the tube number of the PCR Products differs from the tube number of the extraction. The date defaults to the same date as the day the extractions took place but can be edited in the spreadsheet.

Batch Label	Tube Number	Source	Cycle	Forward Prim	Reverse Prim	Date Started	Prepared By	Project	Notes
summer schc	3	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	
summer schc	7	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	
summer schc	5	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	
summer schc	6	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	
summer schc	4	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	
summer schc	2	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	
summer schc	1	Extraction 03	ITS1fITS4-SM 2 ITS1f	1 ITS4	1 ITS4	3/5/25	Miranda Barr	Platypodine Diversity	

PCR PRODUCTS PAGE

The “PCR Products” page stores information about both polymerase chain reaction products and primer stocks used to amplify different regions of DNA.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE PCR PRODUCTS TABLE

The “PCR Products” table includes fields for “Batch Label” (general name for a particular set of PCR products, ideally generated on the same day using the same cycle), the “Tube #” (specifying which PCR product is which within a given batch – traditionally this would be numeric, but the database also accepts text), the “Source” (from which tissue was derived for input), whether or not amplification was a “Success” (indicated by a blue checkmark), the “Cycle” used, the “Forward Primer” used, the “Reverse Primer” used, the “Date [the PCR procedure was] Started”, who the PCR product was “Prepared By”, the “Project” it was prepared for, and any additional “Notes” associated with the PCR product.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

PCR Dashboard				
Batch Label	Tube Numb...	Source		Success
Summer School 03/28/20...	1	Extraction 03/04/2025 summer school 1 from Culture P492 MYEA from P459 Geosmithia Ma, 2025-02-08		✓
Summer School 03/28/20...	2	Extraction 03/04/2025 summer school 2 from Culture P511 MYEA from P455, Ambrosiozyma sp., Ex, 2025-02-18		✓
Summer School 03/28/20...	3	Extraction 03/04/2025 summer school 3 from Culture P504 MYEA from P449, Neocosmospora ambrosium, Ma, 2025-0...		✓
Summer School 03/28/20...	4	Extraction 03/04/2025 summer school 4 from Culture P525 MYEA from P515, Irpex ambrosius Ex, 2025-02-26		✓
Summer School 03/28/20...	5	Extraction 03/04/2025 summer school 5 from Culture P503 MYEA from P462 Ex E. cobbl, 2025-02-14		✓
Summer School 03/28/20...	6	Extraction 03/04/2025 summer school 6 from Culture P508 MYEA from P448, Raffaelea xyleborini, Ex, 2025-02-18		✓
Summer School 03/28/20...	7	Extraction 03/04/2025 summer school 7 from Culture P509 MYEA from P448, Raffaelea xyleborini, Ph, 2025-02-18		✓
399 results				

Different culture fields can be filtered, as in “Projects”. Filtering options are shown below.

Filtering Options

Batch Label

Separate different search terms with semicolon-space

Tube Number

Success Status

☒ Successful
 ☒ Failed or Untested

Date Range

Jan 1, 2019 - May 11, 2025

Source

Separate different search terms with semicolon-space

Cycle

ITS1fITS4-SMITH ✓

ITS_LESF_AR ✓

LSU_AR ✓

PCR53 ✓

Forward Primer

5 Bt2a ✓

8 ITS1 ✓

2 ITS1f ✓

7 ITS5 ✓

Reverse Primer

6 Bt2b ✓

1 ITS4 ✓

4 LR5 ✓

Lab Member

Miranda Barnes ✓

Katy Deitz ✓

Yiyi Dong ✓

Jiri Hulcr ✓

Projects

Euplatypus Project (Dummy Data) ✓

General Lab Use ✓

Platypodine Diversity ✓

SPB Fungal Symbiont Genome ✓

Notes

Separate different search terms with semicolon-space

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Prepared By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited as text by any logged-in user. “Batch Label”, “Tube Number”, “Source”, “Cycle”, “Forward Primer”, “Reverse Primer”, “Prepared By”, and “Project” can be edited through a dropdown menu. “Date Started” can be edited through a calendar menu. Multiple sources may be selected and removed at once for a given table. Users must press save for their changes to be preserved. PCR products can be deleted from the “PCR Products” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “PCR Products” table.

CREATING NEW PCR PRODUCTS

The “PCR Products” input framework is very similar to the one for “Extractions”.

There are two options for filling the input table. The first is to use the “Create New PCR Batch” form. When using this system, users select a quantity for “Number of Tubes” (defaulting to 8 as the standard strip tube length) and provide text input for “Batch Label” and “Notes”. They use dropdown menus to specify the the “Project”, the “Forward Primer” stock used, the “Reverse Primer” stock used, the “Cycle” used, and who it was “Prepared By” (and can also modify the “Date Started” accordingly). When they click “Submit” the “Input Table” will populate with the information they have just provided. Any of the fields (including “Tube Number”) can be edited, but all of source fields **must** be individually specified using a drop-down menu.

Create New PCR Batch

Project *
Select an option

Batch Label *
Label associated with whole PCR batch in fridge

Forward Primer *
Select an option

Reverse Primer *
Select an option

Cycle *
Select an option

Date Started *
May 11, 2025

Prepared By *
Miranda Barnes

Number of Tubes
8

Notes
Enter value

Clear Form

Submit

New Cycle Form

Cycle Name
Enter new cycle name

Definition
Describe run parameters

Submit

Bulk Input By File

Select  CSV file with PCR information

Review Submission/Download Template								
Batch Label	Tube Number	Source	Cycle	Forward Pri...	Reverse Pri...	Date Started	Prepared By	Project
summer school e...	3	Extraction 03/04/2025 summer school 3 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
summer school e...	7	Extraction 03/04/2025 summer school 7 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
summer school e...	5	Extraction 03/04/2025 summer school 5 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
summer school e...	6	Extraction 03/04/2025 summer school 6 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
summer school e...	4	Extraction 03/04/2025 summer school 4 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
summer school e...	2	Extraction 03/04/2025 summer school 2 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
summer school e...	1	Extraction 03/04/2025 summer school 1 fro...	ITS1fITS4-SMITH	2 ITS1f	1 ITS4	Mar 5, 2025	Miranda Barnes	Platypodine D
7 results								
Submit All				Reset Input Table				

download either the raw or mapped data from the “Input Table”, and copy/modify it as needed in a spreadsheet software of the user’s choice. Users can then ensure that their data is properly formatted for entry into the table. Although this is also true for “PCR Products”, the formatting for the source is very specifically “Extraction [Batch Label] [Tube Number] from [Source Type] [Source ID]”, which can be annoying for bulk-populating data. To bypass such problems, please use the “PCR Products from Selected Extractions” form on the “Extractions” page to generate an appropriate .csv file quickly and easily.

Users will be prompted to save any changes made, as in the Projects table. **As with the Projects Table, if users do not save changes made to the Input Table, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This will also have the effect of clearing the Input Table. If the user wants to clear the input table without adding their entries to the database, they can click the “Reset Input Table” button.

As with preceding pages, if there are duplicate “Batch Label” + “Tube Number” combinations, or there is missing information from required fields (i.e., “Batch Label”, “Tube Number”, “Source”, “Cycle”, “Forward Primer”, “Reverse Primer”, “Date Started”, “Prepared By” and “Project”), you will not be able to “Submit All.” Problematic rows will be highlighted in red until you make and save appropriate edits.

Review Submission/Download Template

Batch Label	Tube Number	Source	Cycle	Forward Pri...	Reverse Pri...	Date Started	Prepai
sample batch	1	Extraction 03/04/2025 summer school 3 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	2	Select option	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	3	Extraction 03/04/2025 summer school 2 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	4	Extraction 03/04/2025 summer school 5 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	5	Extraction 03/04/2025 summer school 6 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	6	Extraction 03/04/2025 summer school 6 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	7	Extraction 03/04/2025 summer school 3 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai
sample batch	8	Extraction 03/04/2025 summer school 1 fro...	LSU_AR	3 LROR	4 LR5	Feb 18, 20...	Mirai

8 results

Please correct missing information in indicated rows under "Source".

Submit AllReset Input Table

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

On the “PCR Products” page, there is only one categorical field to which users are permitted to add new options: Cycle. Hence, the “New Options” form is reduced to a “New Cycle” form for the “PCR Products” page. It is otherwise operated identically. Generated terms and definitions are displayed in “Table Terms” at the bottom of the page.

USING THE PRIMER STOCKS TABLE

The “Primer Stocks” table (within the “Primer Stocks” dropdown container) includes columns for “Primer ID”, “Acronym” (i.e., ITS1f, LROR, etc.), the “Direction” the primer runs in, the “Availability” of a particular stock, the “Sequence” of a particular primer, and any other “Notes” on that stock. The idea behind this table is that a user could detail information for primer stocks as a quality control measure in case they need to figure out

whether a particular stock keeps failing or appears to have become contaminated. That said, at present in our lab, it acts more as a list of primers generally than of their stocks.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

Primer Stocks

Primer ...	Acron...	Direction	Availabil...	Sequence	Notes
8	ITS1	Forwa...	✓	TCCGTAGGTGAACCTGCGG	Used
7	ITS5	Forwa...	✓	GGAAGTAAAAGTCGTAACAA...	Used
6	Bt2b	Reverse	✓	ACCCTCAGTGTAGTGACCCT...	Used
5	Bt2a	Forwa...	✓	GGTAACCAAATCGGTGCTGC...	Used
4	LR5	Reverse	✓	TCCTGAGGGAAACTTCG	Used
3	LROR	Forwa...	✓	ACCCGCTGAACCTTAAGC	Used
2	ITS1f	Forwa...	✓	CTTGTCATTAGAGGAAGTAA	Used
8 results					

New Primer

Acronym *

Direction

ITS4, LROR, etc.

Direction of the primer

Target Sequence

Target sequence of primer

Notes

Any additional information about the primer

Submit

Different “Primer Stocks” fields can be filtered, as in “Projects”. Filtering options are shown below.

Filtering Options

Primer ID

Separate different search terms with sem

Direction

☒ Forward

☒ Reverse

Availability

☒ Available

☒ Unavailable

Target Sequence

Separate different search terms with semicolon-space

Acronym

Separate different search terms with sem

Notes

Separate different search terms with semicolon-space

Only “Availability” and “Notes” can be edited in the “Primer Stocks” table, but they can be edited for all entries by any logged-in user. All changes to the table must be saved as described above. Items can be deleted from the “Primer Stocks” table as described for the “Projects” Table unless they are referenced by entries in the “PCR Products” table.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Primer Stocks” table.

CREATING A NEW PRIMER STOCK

To create a new primer stock, users provide text input for “Acronym”, “Target Sequence”, and “Notes” and then select from a dropdown menu for “Direction”. If an acronym that already exists in the table is used for a stock, its target sequence and direction will automatically be populated in their respective fields. Users must provide information for “Acronym”. Click submit to add the entry to the database.

PCR PRODUCTS FROM EXTRACTIONS

When you click to select entries in the “PCR Products” table, you’ll also see a form pop up beneath “Filtering Options” and beside “Create New Batch” offering to help the user create “Sequencing From Selected PCR Products”. This is to speed input on the subsequent “Sequencing” page. Users select a project, device, and

“Date Started”, and they indicate who submitted the PCR product(s) for sequencing and add any additional notes they have on the sample.

Sequencing From Selected PCR Products

Project *
General Lab Use

Device *
Eurofins Sanger Se...

Date Started *
May 11, 2025

Submitted By *
Miranda Barnes

Notes
Enter value

Download Table

When a user presses “Download Table”, they receive a .csv file that can be brought to the subsequent “Sequencing” page as a template for the submission table there. Note that you will still need to specify the output folders for each sequence individually before submission.

Last In-House Sample	Device	Output Folder	Date Started	Submitted By	Project	Notes	Taxon	Accession	% Identity	Date Identified	Identified By	Identification Notes
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	
PCR Product Summer School 03/28/21	Eurofins Sanger Sequencing (Premixed)		2025-05-11T	Miranda Barri	General Lab Use					2025-05-11T	Miranda Barnes	

SEQUENCES PAGE

The “Sequences” page stores information on sequences received back from various sequencing services, as well as mechanisms for assigning identifications to those sequences based on BLAST searches and tree-building. Entries in the “Sequences” table all refer to a “Last In-House Sample” rather than a more direct object (i.e., a library). This is to acknowledge that in our lab, both extractions and PCR products may be sent off for sequencing, and we do not prepare our own libraries.

USING THE SEQUENCES TABLE

The “Sequences” table includes fields for “Accession” (showing the accession number of the sequence to which it shares the closest similarity to in the BLAST database), the “taxon” assigned to it based on sequence analyses, a “Last In-House Sample” (displaying the item that was sent in for sequencing), the “Device” (or platform used for sequencing), the “Output Folder” (i.e., where the sequence information is stored either on Dropbox or in our lab drive), the “Date Started” reflecting when the sample was submitted, the person the sample was “Submitted By”, the “Project” it was submitted for, and any additional “Notes” on the sequence or sequencing output.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in Projects. Concealed columns can be revealed by scrolling to the side.

Sequences Dashboard					
Accession	Taxon	Last In-House Sample	Device	Output Folder	
	Entomocorticium cobbii	Extractions entocheck6/24 1 Culture P177 MEA CP from S18273, 2024-05-23	Eurofins Sanger Sequencing (Premix...	Lab Server F	
		PCR Product fromfrozenSPB5/24 1 Culture P165 BSMA from BV23216 (frozen myc), 2024-05-10...	Eurofins Sanger Sequencing (Premix...	Lab Server F	
		PCR Product fromfrozenSPB5/24 3 Culture P166 BSMA from BV23216 (frozen myc), 2024-05-10...	Eurofins Sanger Sequencing (Premix...	Lab Server F	
		PCR Product fromfrozenSPB5/24 5 Culture P167 BSMA from BV23216 (frozen myc), 2024-05-10...	Eurofins Sanger Sequencing (Premix...	Lab Server F	
	Graphilbum puerense	PCR Product entotestcov4/24 3 Culture P148 MYEA CP from CP139, 2024-03-29	Eurofins Sanger Sequencing (Premix...	Lab Server F	
5 results					

Different “Sequences” fields can be filtered, as in Projects. Filtering options are shown below.

Accession Number

Separate different search terms with semicolon-space

Taxon

Separate different search terms with semicolon-space

Derived From

Separate different search terms with semicolon-space

Device

Eurofins Sanger Sequencing (Premixed) ✓

Output Folder

Separate values with semicolon-space

Date Range

Jan 1, 2019 - May 12, 2025

Lab Member

Miranda Barnes ✓

Katy Deitz ✓

Yiyi Dong ✓

Jiri Hulcr ✓

Notes

Separate different search terms with semicolon-space

Projects

euplatypus Project (dummy data)

General Lab Use ✓

Platypodine Diversity ✓

SPB Fungal Symbiont Genome ✓

SPB Genome ✓

SPB Mycangial Transcriptome ✓

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Prepared By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited as text by any logged-in user. As with the “Beetles” table, accession and taxon cannot be edited directly through this central table and should be added and modified separately in the “Identifications” container. “Last In-House Sample”, “Device”, “Prepared By”, and “Project” can be edited through a dropdown menu; “Output Folder” can be edited as text. “Date Started” is edited via a calendar menu. Users must press save for their changes to be preserved. Sequences can be deleted from the “Sequences” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream.

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Sequences” table.

USING THE IDENTIFICATIONS TABLE

The “Identifications” table includes fields for the unique “Accession Number”, the “Taxon” to which the sequence has been assigned, the “% Identity” that a sequence shares with a BLAST hit used in making that identification, the “Output Folder” where the data currently resides, the “Date Identified”, the person it was “Identified By”, and any “Notes” on the identification. This table exists separately from the “Sequences” table for a few reasons. Firstly, this allows multiple identifications to be applied to a given sequence via different means (for example, an identification made based on close identity in a BLAST search versus identifications made based on placements in a tree). Secondly, it divorces the act of identification from the actual act of sequencing, which are two distinct steps of a given pipeline.

To view the identifications assigned to different sequences, users click on the checkboxes beside the sequences they want to examine. The “Make New Identification” form will also automatically populate with the ID for the most recently selected sequence. Identifications can be deleted from the “Identifications” table in the same way that they can from the “Sequences” table (also restricted by user). However, it is recommended that users retain even erroneous identifications to maintain an accurate history of how a sample was treated over the course of its lifespan. You cannot edit identifications entries for this reason – if you want to add a new taxon to the table for a sample, you must create a new entry.

Identifications						
Accessi...	Taxon	% Identi...	Output Fold...	Date Identified	Identified By	Notes
	Raffaelea xyleborini	0%		Aug 29, 2024 10:29 PM	Andre Rodrigu...	Enter value
	Starmera sp.	0%		Aug 29, 2024 10:29 PM	Andre Rodrigu...	Enter value
210 results						

Identifications can be downloaded and filtered as with the “Projects” table using the buttons at the base of the display. Filtering options are shown below

Filtering Options		
Accession Number Separate different search terms with semicolon-space	Output Folder Separate values with semicolon-space	
Taxon Separate different search terms with semicolon-space	Date Range Jan 1, 2019 - May 12, 2025	Lab Member Miranda Barnes ✓ Katy Deitz
% Identity 0-100		
Notes Separate different search terms with semicolon-space		

CREATING A NEW IDENTIFICATION

To make a new identification, users must select a “Sequence” (formatted as “[the source] in [the output folder]”), add a taxon, indicate the date identified, record who identified the beetle, and optionally include information about the accession number, percent identity, and any additional notes. Users provide text input for “Taxon”, “Accession Number”, and “Notes”. They may select from a dropdown menu for “Sequence”,

and “Identified By”, “% identity” is modified numerically and “Date Identified” can be changed via a calendar menu. Everything else is input as text. To create a new identification, users must click submit.

Make New Identification

Source for Sequencing *

Extractions entocheck6/24 1 Culture P177 MEA CP from S18273, 2024-05-23 in Lab Server Files Backup / Miranda Barnes / Genome Project / EntCheckEurofins (https://www.dropbox.com/scl/...

Taxon *

Species name

Accession Number

Add accession number (optional, can be added later if necess

% identity

0.0%

Date Identified *

May 12, 2025 9:17 AM

Identified By *

Miranda Barnes

Notes

Enter value

Submit

CREATING A NEW SEQUENCE

There are a few ways to generate new sequences. To create a single new “Sequences” entry, users provide text input for “Output Folder File Path” and “Notes”. They then select from dropdown menus for the “Last In-House Sample”, “Device”, “Submitted By”, and “Project”. Users must provide information for all fields except for “Notes” and “Output Folder File Path”. Selecting a “Project” populates subsequent entries in the dropdown menu “Last In-House Sample”.

Users providing file paths should ideally be as detailed as possible so that other users can locate their item. If they include a link to where the file is stored on Dropbox, this is even better than just providing a simple list of nested folders.

New Sequencing Event

Project *

Select an option

Last In-House Sample *

Select options

Device *

Select an option

Date Started *

Feb 18, 2026

Submitted By *

Miranda Barnes

Accession Number

Add accession number (optional, can be added later if necessary)

Taxon

Species name

% identity

Enter value

Output Folder File Path

Complete file path to output information

Notes

Enter any additional information here

Clear Form

Submit

New Device Form

Device Name

Enter device name

Definition

Description of device

Submit

Bulk Input By File

Select  CSV file with sequencing information

Users can also bulk upload samples using the “Bulk Input By File” option. Uploading files in this way or pressing submit on the single-entry form will populate a “Review Submission” table that can be examined prior to submission. As with the proceeding pages, our recommendation is to download material from the review table generated from single entries as a table and then use a spreadsheet program to replicate and edit as much information for bulk submissions as necessary.

Review Submission/Download Template

Accession	Taxon	Last In-House Sample	Output Folder	Device	% Ident...
Enter value	Enter va...	Extractions 03/04/2025 summer school 1 Culture P492 MYEA from P459 Geosmithia Ma, 2025-02...	sample path ...	Eurofins Sanger Sequencing (Premix...	Enter v...

1 result

Reset Input Table
Submit All

Users will be prompted to save any changes made, as in the Projects table. **As with the Projects Table, if users do not save changes made to the Input Table, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This will also have the effect of clearing the Input Table. If the user wants to clear the input table without adding their entries to the database, they can click the “Reset Input Table” button.

As with preceding pages, if there is missing information from required fields (i.e., “Last In-House Sample”, “Device”, “Date Started”, “Prepared By” and “Project”, or, if you include information in “Taxon”, also “Date Identified” and “Identified By”), you will not be able to “Submit All.” Problematic rows will be highlighted in red until you make and save appropriate edits.

Review Submission/Download Template

Accession	Taxon	Last In-House Sample	Output Folder	Device	% Id
Enter value	Enter va...	Extractions 03/04/2025 summer school 2 Culture P511 MYEA from P455, Ambrosiozyma sp., Ex, 2...	/sample/out...	Select option	Ente

1 result

Please correct missing information in indicated rows under "Device".

Submit All
Reset Input Table

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

On the “Sequences” page, there is only one categorical field to which users are permitted to add new options: Device. Hence, the “New Options” form is reduced to a “New Device” form for the “Sequences” page. It is otherwise operated identically. Generated terms and definitions are displayed in “Table Terms” at the bottom of the page.

PROCESSES PAGE

The “Processes” page stores information on miscellaneous lab procedures (particularly those producing documents) where the ability to link information to other entries is important, but having tables to track each of these individually may generate unnecessary complexity. These currently include pictures from gel electrophoresis, Tapestation results, etc., but this could easily be extended to other protocols not currently reflected in the database structure.

Users must log in before they are able to add or edit new entries. The process of logging in is as described for “Projects”.

USING THE PROCESSES TABLE

The “Processes” table includes fields for “File Path/Link to Output” (describing where the document from the process is located), “Associated Object(s)” (the samples the procedure were performed on), the “Pipeline” used to generate the data, the “Date Created”, the person the sample was “Analyzed By”, the “Project” it was analyzed for, and any additional “Notes” on the process or process output.

The dimensions of the columns in the table can be altered by clicking and dragging the column edges at the top of the table where the headings sit, as in “Projects”. Concealed columns can be revealed by scrolling to the side.

Processes Dashboard					
	Pipeline	Date Creat...	Analyzed By	Project	Not...
	Tapestation	Sep 30, 2024	Miranda Barn...	X. crass Genome	
	Tapestation	Sep 20, 2024	Miranda Barn...	X. crass Genome	
	Qubit and Nanodrop DNA ...	Sep 19, 2024	Miranda Barn...	X. crass Genome	
from Culture P284 PDA from CP	Qubit and Nanodrop DNA ...	Sep 12, 2024	Miranda Barn...	SPB Fungal Symbiont Genome	
ractions 8/29 Pool to Column Tu	Qubit and Nanodrop DNA ...	Aug 29, 2024	Miranda Barn...	X. crass Genome	
46 results					

Different “Processes” fields can be filtered, as in “Projects”. Filtering options are shown below.

Filtering Options

File Path/Link to Output

Separate different search terms with semicolon-space

Associated Objects

Separate different search terms with semicolon-space

Pipeline

Gel Electrophoresis Imaging

Qubit and Nanodrop DNA QC

Tapestation

Date Range

Jan 1, 2019 - May 12, 2025

Notes

Separate different search terms with semicolon-space

Lab Member

Miranda Barnes

Katy Deitz

Projects

General Lab Use

X. crassiusculus Fungal Symbiont Genome

As with the “Beetles” table, entries can only be edited if they’re associated with your lab member identity (i.e., the person listed in the “Analyzed By” column is the same as the one who is currently logged in). The only exception to this rule is “Notes”, which can be edited as text by any logged-in user. “File Path/Link to Output” can also be edited as text. “Associated Object(s)”, “Pipeline”, “Analyzed By”, and “Project” can be edited through a dropdown menu, and “Date Created” can be edited via a calendar menu. Users must press save for their changes to be preserved. Sequences can be deleted from the “Processes” table as described for the “Beetles” Table – i.e., only if an entry is associated with the person who is logged in and isn’t referenced downstream (in this case, usually only by other “Processes”).

The same rules about raw and human-readable downloads that applied to the “Projects” table also apply to the “Processes” table.

CREATING A NEW PROCESS ENTRY

To create a new “Processes” entry, users provide text input for “File Path/Link to Output” and “Notes” and then select from dropdown menus for “Analysis Pipeline”, “Associated Object(s)”, “Analyzed By”, and “Project”. Users must provide information for all fields except for “Notes” and “Analysis Pipeline”. Selecting a “Project” will populate the dropdown menu for “Associated Object(s)” with potential entries that a user can choose from for that project. Users can select multiple options for this field.

Users providing file paths for “File Path/Link to Output” should ideally be as detailed as possible so that other users can locate their item. If they include a link to where the file is stored on Dropbox, this is even better.

New Process Entry

Project *

Select an option

Date Started *

May 12, 2025

Analyzed By *

Miranda Barnes

Associated Object(s) *

Select an option

Analysis Pipeline

Select an option

File Path/Link to Output *

Where the output of the analysis can be found on Dropbox

Notes

Any additional information about this entry

Submit

New Pipeline Option

Pipeline Name

Shorthand for pipeline

Full Process

Steps taken in the pipeline (link to GitHub/shared doc)

Submit

Users can also bulk upload samples using the “Bulk Input By File” option. Uploading files in this way or pressing submit on the single-entry form will populate a “Review Submission” table that can be examined prior to submission. As with the proceeding pages, our recommendation is to download material from the review table generated from single entries as a table and then use a spreadsheet program to replicate and edit as much information for bulk submissions as necessary.

Bulk Input By File

Select CSV file with process information

Review Submission/Download Template

File Path/Link to Output	Associated ...	Pipeline	Date Creat...	Analyzed By	Project	Notes
/sample/file/output	Extractio...	Tapestation	Feb 18, 20...	Miranda Barn...	SPB Mycangial Tra...	Enter value

1 result

Submit All

Reset Review Table

Users will be prompted to save any changes made, as in the Projects table. **As with the Projects Table, if users do not save changes made to the Input Table, they will not be retained. If users press “Submit All” without saving, their changes will not be reflected in the information provided to the database.**

“Submit All” sequentially inserts rows corresponding to each entry into appropriate tables in the database. This will also have the effect of clearing the Input Table. If the user wants to clear the input table without adding their entries to the database, they can click the “Reset Input Table” button.

As with preceding pages, if there is missing information from required fields (i.e., “Associated Object(s)”, “Pipeline”, “Date Created”, “Analyzed By” and “Project”), you will not be able to “Submit All.” Problematic rows will be highlighted in red until you make and save appropriate edits.

Review Submission/Download Template

File Path/Link to Output	Associated Object(s)	Pipeline	Date Creat...	Analyzed By	Project	Notes
/sample/file/output	Extractions SPB Prothorax 2025...	Select option	Feb 18, 20...	Miranda Barn...	SPB Mycangial Tra...	Enter value

1 result

Please correct missing information in indicated rows under "Pipeline".

Submit All

Reset Review Table

CREATING NEW OPTIONS FOR CATEGORICAL FIELDS

On the “Processes” page, there is only one categorical field to which users are permitted to add new options: “Pipeline”. Hence, the “New Options” form is reduced to a “New Pipeline” form for the “Processes” page. It is otherwise operated in the same way as all the previous pages. Generated terms and definitions are displayed in “Table Terms” at the bottom of the page. As with “New Method” on the “Extractions” page, it may not be convenient to specify a pipeline in as much detail as would be helpful to other lab members in the term and definition fields. Instead, users are encouraged to provide links to documents or webpages containing a more detailed summary of the pipeline in the definition rather than providing vague descriptions of what they mean by a given procedure.

SAMPLE TRACKING PAGE

The last page listed in the navigation bar at the top of the screen is the “Sample Tracking” page. This part of the application does not permit any data entry and functions only to holistically monitor information associated with various objects in the database.

To begin, users should decide which “Projects”, “Lab Members”, and “Date Range” they want to display samples over using the menu at the top of the page. The default setting encompasses all lab members and the entire date range over which samples exist. Clicking and unclicking or modifying the range listed on these filters allows users to narrow down the objects they are looking for/at. **For ideal functionality, select only one or two projects at a time. Do not select all projects from the Projects dropdown list unless other filters substantially limit the possible number of samples involved.** Different selections can be removed or modified by pressing the “Clear Projects”, “Clear Members”, and “Clear Date Range” options.

Sample Tracking

Projects

General Lab Use
X. crassiusculus Fungal Symbiont Genome
SPB Fungal Symbiont Genome
Wingfield Sabbatical

Select a Project

Lab Member

Miranda Barnes ✓
Katy Deitz ✓
Yiyi Dong ✓
Jiri Hulcr ✓

Date Range

Jan 1, 2019 - May 12, 2025
Clear Projects
Clear Members
Clear Date Range

Once a desired frame has been established, tabs will appear corresponding to tables that have data associated with the criterion provided.

Beetles Cultures Slants Extractions PCR Sequencing

Download All Tables

Clicking on each tab, you can see tables related to each sample type. These displays are identical to the tables on their corresponding dashboards except that information cannot be added or modified.

Beetles Cultures Slants Extractions PCR Sequencing

Beetle ID	Taxon	Life Sta...	Sex	Sample Si...	Sampling Protocol	Derived From	Vial Numb...	Date Collect...	Collected ...
37773	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37772	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37771	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37770	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37769	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37768	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37766	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...
37765	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...

36 results

Download All Tables

When users select a sample or multiple samples in one of these tables, objects in the pipeline that did not either precede each sample (i.e., the sample was derived from that object, derived from an object derived from that object, and so on) or follow each sample (i.e., the object was derived from the sample, derived from an object that was derived from the sample, and so on) are removed from the other tables.

Users can choose to select multiple samples within a table at once.

Beetles Cultures Slants Extractions PCR Sequencing										
Beetle ID	Taxon	Life Sta...	Sex	Sample SI...	Sampling Protocol	Derived From	Vial Numb...	Date Collect...	Collected ...	
37773	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
37772	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
✓ 37771	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
✓ 37770	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
37769	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
37768	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
37766	Euplatypus p...	Adult	Male	1	Flying or trapped			Jun 16, 2022	Andre R...	
2 of 36 selected										

Beetles Cultures Slants Extractions PCR Sequencing										
Cultures										
Physical La...	Me...	Source	Tissue	Technique	Date Creat...	Prepared By	Project	Incubation Temperat...	Fridge Date	D
21468	19	Beetle 37770	head	dilution-1	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
21469	19	Beetle 37770	head	dilution-01	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
21470	19	Beetle 37770	prothorax	dilution-1	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
21471	19	Beetle 37770	prothorax	dilution-01	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
21472	19	Beetle 37771	hindgut	dilution-1	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
21473	19	Beetle 37771	hindgut	dilution-01	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
21474	19	Beetle 37771	head	dilution-1	Jun 19, 20...	Andre R...	Euplatypus Projec...	25		2
19 results										

Before looking at a new sample derivation pathway, deselect other entries and wait for the tables to reset before proceeding. Also, watch the bottom of the page (where ReTool indicates queries are running) while examining the outcome of selections to ensure that all associated queries required to track samples back through the database have been completed before making any inferences based on the displayed information (example shown below – see “Running culturesFiltered: 2.3s” shows that the culturesFiltered query is still being retrieved from the database).

Beetles

Cultures

Extractions

PCR

Sequencing

Retool

production

main

Latest

Accessi...

Taxon

Last In-House Sample

Device

MT629...

Dryadomyces amasa...

PCR Product OXOPLAT-2025-08-19 1578 Culture P1578 SMA 2 from P1433 2025-08-01

Eurofins Sanger Sequencing (Premix...

PP1257...

[Candida] sp. (uncert...

PCR Product OXOPLAT-2025-08-20 1659 Culture P1659 SPDA 8 from P1391 2025-08-05

Eurofins Sanger Sequencing (Premix...

NG_055...

Ambrosiozyma ambro...

PCR Product OXOPLAT-2025-08-20 1624 Culture P1624 SPDA 5 from P1472 2025-08-05

Eurofins Sanger Sequencing (Premix...

KX7533...

Brettanomyces acido...

PCR Product OXOPLAT-2025-08-19 1598 Culture P1598 SMYEA 1 from P1361 2025-08-01

Eurofins Sanger Sequencing (Premix...

LC3635...

Raffaelea cyclorhipidi...

PCR Product OXOPLAT-2025-08-19 1673 Culture P1673 SPDA 8 from P1466, 2025-08-09

Eurofins Sanger Sequencing (Premix...

NG_055...

Ambrosiozyma ambro...

PCR Product OXOPLAT-2025-08-20 1543 Culture P1543 CSPDA 5 from P1364 2025-08-01

Eurofins Sanger Sequencing (Premix...

MT629...

Dryadomyces amasa...

PCR Product OXOPLAT-2025-08-19 1607 Culture P1607 SPDA 2 from P1448 2025-08-05

Eurofins Sanger Sequencing (Premix...

LC3635...

Raffaelea sp. persona...

PCR Product OXOPLAT-2025-08-19 1602 Culture P1602 SMYEA 8 from P1361 2025-08-01

Eurofins Sanger Sequencing (Premix...

Ambrosiozyma sp. Iso...

PCR Product OXOPLAT-2025-08-19 1648 Culture P1648 SPDA 8 from P1401 2025-08-05

Eurofins Sanger Sequencing (Premix...

61 of 61 selected

Running culturesFiltered: 2,3s

Debug

Users can download tables individually using the download icon at the bottom left-hand corner of each table, or they can use the “Download All Tables” button beneath the tabbed container to download all tables at once in a .zip file.

GENERATING DATA TABLES AND MATRICES

When the user navigates to the “Sequencing” tab, they will find a form consisting of several checkboxes for each table, followed by a button to “Prepare Sample Table”. “Primary Isolation” is defined as any culture associated with an object that is not derived from an other culture; “Subcultures” are any subsequent cultures from that culture. “BLAST Taxon” refers to the highest percent-identity identification that has a BLAST accession number associated with it; “Current Taxon ID” is whatever the most recent identification assigned to the sample is (i.e., as inferred from examining a phylogenetic tree, for example).

Plants
☐ Plant Label ☐ Taxon ☐ Health
☐ Date Recorded ☐ Recorded By
☐ Project ☐ Notes

Galleries
☐ Gallery ID
☐ Part Sampled
☐ Date Recorded
☐ Recorded By
☐ Project ☐ Notes

Traps
☐ Trap ID ☐ Trap Type ☐ Lure
☐ Substrate ☐ Date Created ☐ Set By
☐ Project ☐ Notes

Beetles
☐ Beetle ID ☐ Taxon ☐ Life Stage ☐ Sex
☐ Sample Size ☐ Sampling Protocol ☐ Vial Number
☐ Country ☐ State/Province ☐ County ☐ Locality
☐ Latitude ☐ Longitude ☐ Date Collected
☐ Prepared By ☐ Project ☐ Notes

Cultures - Primary Isolation
☐ Physical Label
☐ Morphotypes ☐ Media
☐ Tissue ☐ Technique
☐ Incubation Temperature
☐ Date Created
☐ Fridge Date
☐ Discard Date
☐ Prepared By ☐ Project
☐ Notes

Cultures - Subcultures
☐ Physical Label
☐ Media ☐ Tissue
☐ Technique
☐ Incubation Temperature
☐ Date Created
☐ Fridge Date
☐ Discard Date
☐ Prepared By ☐ Project
☐ Notes

Slants
☐ Slant ID ☐ Media
☐ Location
☐ Storage Method
☐ Date Created
☐ Prepared By ☐ Project
☐ Notes

Extractions
☐ Batch Label
☐ Tube Number
☐ Method
☐ Date Started
☐ Prepared By ☐ Project
☐ Notes

PCR
☐ Batch Label
☐ Tube Number ☐ Cycle
☐ Forward Primer
☐ Reverse Primer
☐ Date Started
☐ Created By ☐ Project
☐ Notes

Sequences
☐ Current Taxon ID
☐ BLAST Taxon
☐ Accession
☐ Percent Identity
☐ Device
☐ Output Folder
☐ Date Started
☐ Prepared By ☐ Project
☐ Notes

Prepare Sample Table

The purpose of this form is to generate a) a .csv file linearizing sample pathways for easy addition to publications and b) occurrence and frequency matrices demonstrating how many times a particular fungal sequence occurs with a particular beetle sample.

To use this feature, select all the sequencing entries you want to be considered in your final output.

Beetles

Cultures

Slants

Extractions

PCR

Sequencing

Accession...	Taxon	Last In-House Sample	Device	Output Folder
☒	Esteya floridana	PCR Product compiledEuplatypus 8685 Culture 21646	Eurofins Sanger Sequencing (Premix...	
☒	Esteya floridana	PCR Product compiledEuplatypus 8687 Culture 21648	Eurofins Sanger Sequencing (Premix...	
☒	Ceratocystopsis lunata	PCR Product compiledEuplatypus 8688 Culture 21659	Eurofins Sanger Sequencing (Premix...	
☒	Raffaalea xyleborini	PCR Product compiledEuplatypus 8691 Culture 21604	Eurofins Sanger Sequencing (Premix...	
☒	Ambrosiozyma sp.	PCR Product compiledEuplatypus 8692 Culture 21660	Eurofins Sanger Sequencing (Premix...	
☒	Raffaalea xyleborini	PCR Product compiledEuplatypus 8693 Culture 21605	Eurofins Sanger Sequencing (Premix...	
☒	Raffaalea xyleborini	PCR Product compiledEuplatypus 8694 Culture 21639	Eurofins Sanger Sequencing (Premix...	
☒	Raffaalea xyleborini	PCR Product compiledEuplatypus 8695 Culture 21610	Eurofins Sanger Sequencing (Premix...	
☒	Raffaalea xyleborini	PCR Product compiledEuplatypus 8696 Culture 21625	Eurofins Sanger Sequencing (Premix...	

176 of 176 selected

Then select all the fields you want to appear in your linearized sample pathway and click “Prepare Sample Table.” By default, the checkboxes only display for sample types that exist in your pipeline (with a few suggested default selection parameters for common columns). For example, in the below example, we see that we have no information for Plants, Galleries, Traps, or Slants, so it is not an option to select fields in these tables.

Cultures - Primary Isolation ☒ Physical Label ☐ Morphotypes ☐ Media ☐ Tissue ☐ Technique ☐ Incubation Temperature ☐ Date Created ☐ Fridge Date ☐ Discard Date ☐ Prepared By ☐ Project ☐ Notes	Cultures - Subcultures ☒ Physical Label ☐ Media ☐ Tissue ☐ Technique ☐ Incubation Temperature ☐ Date Created ☐ Fridge Date ☐ Discard Date ☐ Prepared By ☐ Project ☐ Notes	Extractions ☒ Batch Label ☒ Tube Number ☐ Method ☐ Date Started ☐ Prepared By ☐ Project ☐ Notes	Beetles ☒ Beetle ID ☒ Taxon ☐ Life Stage ☐ Sex ☐ Sample Size ☐ Sampling Protocol ☐ Vial Number ☐ Country ☐ State/Province ☐ County ☒ Locality ☐ Latitude ☐ Longitude ☐ Date Collected ☐ Prepared By ☐ Project ☐ Notes	PCR ☒ Batch Label ☒ Tube Number ☐ Cycle ☒ Forward Primer ☒ Reverse Primer ☐ Date Started ☐ Created By ☐ Project ☐ Notes	Sequences ☒ Current Taxon ID ☒ Blast Taxon ☒ Accession ☒ Percent Identity ☐ Device ☐ Output Folder ☐ Date Started ☐ Prepared By ☐ Project ☐ Notes
--	--	---	--	---	---